

Adaptive decision-tree caps and the exact certification floor for the pseudo-solution pipeline of $\neg\text{PHP}_n$ at characteristic 2: the answer channel, depletion bounds, the AND-query no-lift, the certification floor, the adjacency certificates, and the corrected two-phase tree

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Abstract

We study the pseudo-solution pipeline $\Omega(n, d)$ of Krajíček [8] for the negated pigeonhole principle over $\mathbb{F}_2$, the reduction target of the AC[2]-Frege lower-bound problem. We prove an exact answer-channel law: killed pairs answer deterministically, free pairs answer fair bits marginally and jointly on every row-incomplete query set, a fully queried free row is *parity-locked* (answer XOR 1; only 2^{2d-1} of the 2^{2d} patterns occur), and the row sums answer the determined 0 on every pigeon. A kernel-level verification proves $V(n, d)^\rho = V(2d, d)$ exactly: one design space, the outer/canonical ambiguity dissolves, the printed vanishing condition itself forces the row parities, and the i.i.d. law rests on the zeroing completion (the earlier disjoint-support route fails at $d \geq 2$). On this law we prove a hypergeometric depletion bound via negative association; exact success caps for fixed-label trees, for non-adaptive variable- and monomial-class trees, and for all adaptive single-variable trees within their budget; and a no-lift theorem: the per-hit freeness posterior of a degree-2 AND query equals, by an algebraic identity, the single-variable Bayes posterior. In the coin model, our main theorem settles the unbounded-budget certification problem at degree 1: every adaptive degree-1 tree errs with probability at least $\text{err}^* = (1 - 2d/n)(2d + 1)(2d)! / 2^{(2d+1)2d} = 2^{-\Theta(d^2)}$, attained by a non-adaptive full-scan counting tree, through sound line-count certificates and a near-permutation obstruction; under the pipeline’s parity-locking the order survives and the exact constant is open. An earlier draft asserted an exact law $1 - 2^{-(2d+1)}$ for the two-phase certification tree; we retract it as a simulator artifact (the simulator certified freeness on a row-XOR condition that no degree-1 query implements), and against the literal pipeline the tree is dead outright. The per-hit caps are sharp but not per-tree: two adjacent 1s certify a free pair *with certainty* (a matched pair forces its row-neighbors to 0), and a scan-then-confirm chain, sound on the literal pipeline, measures success 0.9317 at budget 1000 at both

*Draft prepared for the corpus owner; not submitted and not peer-reviewed. The pipeline $\Omega(n, d)$, the solution condition (Definition 2.2), and the chi-task (Theorem 2.3) are due to J. Krajíček [8], building on [2]; they are quoted here from the printed definitions of [8]. All remaining results (Theorems 3.2, 5.3, 6.1, 7.1, 8.7, Propositions 4.3, 4.4, 4.6, 3.4, 8.9, and Lemmas 3.1, 3.6, 4.1, 4.2, 8.2, 8.4, 8.5, 8.6, 9.1) are original to this research session. This revision retracts the earlier exact two-phase law and its optimality conjecture (a simulator artifact; Remark 7.2) and the all-degrees non-adaptive cap (Remark 4.7); the final revision integrates the kernel-level verification (one design space, parity-locked full rows), the adjacency certificates (Lemma 9.1) and the confirmation-chain measurements, the Theorem-8.7 parity-locking remark, and the theorem-faithfulness table (Table 3).

(32, 2) and (64, 4) (0.9417 on the exact pipeline at (32, 2)) with posterior 1.0000 on every certified output, the certificate rate following $1 - e^{-B/400}$; the scaling of that constant in (n, d) is the sharpest open problem. We tabulate which results are kernel-faithful (the caps, the posteriors, the completion, both surviving certificate mechanisms) and which are coin-model-only (the retracted two-phase law, the floor’s constant, the AND identity’s pipeline transfer). Every attacker remains over the printed budget $e' = d \log k$: the open problem is the *budgeted* certification floor over the certificate trichotomy — parity (dead), counting (inert within budget), adjacency (alive) — with the chi-hypothesis open in exactly its printed budgeted form, together with the odd-characteristic analogue.

1 Introduction

1.1 The proof-complexity target

By the Cook–Reckhow theorem [3], a propositional proof system with polynomially bounded proof size exists if and only if $\text{NP} = \text{coNP}$; hence any super-polynomial lower bound for a sufficiently strong propositional proof system implies $\text{NP} \neq \text{coNP}$ and therefore $\text{P} \neq \text{NP}$. For resolution the pigeonhole principle has exponential lower bounds [5], and so it has for constant-depth Frege [1]. Adding MOD_p gates defeats the Razborov–Smolensky circuit technology [11], and no non-trivial lower bound is known for $\text{AC}[2]$ -Frege on any formula sequence. In 2026 the adjacent frontier moved: an exponential lower bound was proved, with a Lean 4 formalization, for the bit pigeonhole principle in unrestricted dag-like resolution over parities ($\text{Res}(\oplus)$) [10], the system immediately below $\text{AC}[2]$ -Frege, and super-linear lower bounds for random 3-CNFs became available in bounded-depth Frege [9]. The $\text{AC}[2]$ -Frege lower bound for $\neg\text{PHP}_n$ itself is the frontier open problem, and the pseudo-solution program of [8] is currently its most concrete reduction: it replaces the qualitative lower-bound question by a quantitative probabilistic task about explicitly given objects (Theorem 2.3). Proving anything about that task requires, first, an exact understanding of what an adaptive decision tree with low-degree queries can extract from the pipeline $\omega = (\rho, L)$ at the Boolean-relevant characteristic $p = 2$. This paper supplies that understanding for the single-variable, non-adaptive, and single-pass product query classes; proves the answer-channel law at kernel level (one design space, $V(n, d)^\rho = V(2d, d)$, with parity-locked full rows and the i.i.d. law resting on the zeroing completion); settles the unbounded-budget certification problem at degree 1 in the coin model exactly (Theorem 8.7: the optimal error is $2^{-\Theta(d^2)}$, attained by a non-adaptive counting tree; under the pipeline’s parity-locking the order survives and the constant is open); proves the adjacency-certificate lemma and reports the confirmation-chain measurements that beat the per-query caps on the literal pipeline (Section 9); restates the two-phase certification tree under the literal query semantics, after retracting its earlier exact law as a simulator artifact, and shows it dead against the pipeline; reconciles the unbounded results with the printed budget of Theorem 2.3; and isolates the budgeted certification floor over the certificate trichotomy as the precise remaining open problem.

1.2 Contributions

All results concern the pipeline $\Omega(n, d)$ of [8] at $p = 2$ with $n_\rho = 2d$ (Definition 2.1).

1. **The answer channel law, restated at kernel level** (Theorem 3.2): matched pairs answer the constant 1, killed–unmatched pairs the constant 0, free pairs answer fair bits

marginally and jointly on every *row-incomplete* query set, a fully queried free row is *parity-locked* (Theorem 3.2(iv)), and the row sums answer the determined 0 on every pigeon (Proposition 3.4), so aggregate row-sum queries carry no signal at all. The joint law rests on the zeroing completion (Lemma 3.6); the disjoint-support route recorded in an earlier draft fails at $d \geq 2$ (Remark 3.3). A kernel-level verification (Remark 3.5) proves $V(n, d)^\rho = V(2d, d)$ exactly: the outer/canonical design-space ambiguity of the previous draft dissolves, one design space carries the printed vanishing condition, and no reading of the printed definition realizes the coin channel beyond its row-incomplete single-variable law (Corollary 3.8).

2. **Completion of partial patterns** (Lemma 3.6): every partial assignment with fewer than $2d$ pinned coordinates in each row extends to a design, and the number of extensions is independent of the assigned values; a fully pinned row must carry parity 1. Consequently the answer channel of any set of single-variable queries coincides, in joint law, with an i.i.d.-coin channel except on the event that some row's free pairs are queried in their entirety (Corollary 3.8); on that event the row is parity-locked (Theorem 3.2(iv)).
3. **Depletion** (Lemma 4.2): conditioned on any transcript, the freeness probability of any pair whose own cell was not probed exceeds the base rate by a factor $1 + O(s/n)$ at most. This is the hypergeometric negative-association bound, placed on record with the standard citations [6, 4].
4. **Caps**: Proposition 4.3 (fixed-label trees succeed at the base rate f exactly); Proposition 4.4 (non-adaptive single-variable trees with answer-seeking labels succeed at most $f(1 + s/2) + o(1)$); Proposition 4.6 (non-adaptive trees whose queries are single variables or monomials with at most γ distinct variable pairs each succeed at most $(\gamma s + 1)f + o(1)$ at budget s , with the transition $d = \Theta_\gamma(n^{2/3}(\log k)^{-1/3})$; the earlier all-queries generality of this cap is *retracted*, with a counterexample, in Remark 4.7); Theorem 5.3 (all adaptive single-variable trees of depth e succeed at most $\max(q, f)(1 + O(e/n)) + o(1)$, where $q = \frac{2d^2+d}{2d^2-d+n}$ is the exact per-hit Bayes posterior; in particular the success is at most $\frac{1}{2} + o(1)$ for $2d \leq \sqrt{2n}$; the cap is a *budgeted* statement, Remark 5.4). Hence the chi-task hypothesis of Theorem 2.3 holds for these classes with margin.
5. **The AND-query no-lift** (Theorem 6.1): a degree-2 product query is a joint test with no per-hit advantage: its labeling posterior equals q exactly, by the polynomial identity $(f + 2m)^2 = f^2 + 4fm + 4m^2$.
6. **The exact certification floor at degree 1** (Theorem 8.7): over *all* adaptive degree-1 trees with no query budget, the optimal error in the coin model of Definition 8.1 is exactly $\text{err}^* = (1 - 2d/n)(2d + 1)(2d)!/2^{(2d+1)2d} = 2^{-\Theta(d^2)}$, attained by a non-adaptive full-scan counting tree. The proof: linearity of degree-1 answers (Lemma 8.2), full-scan dominance reducing adaptivity to the Bayes problem on the answer table (Corollary 8.3), sound line-count certificates (Lemma 8.4), pigeonhole rescue (Lemma 8.5), and the near-permutation class as the unique obstruction, with pair posterior flat at $2d/n$ (Lemma 8.6). Verification: an exhaustive Bayes audit at $(4, 1)$ and exact enumeration of all 2^{20} coin matrices at $(5, 2)$ (Remark 8.8). The rows-and-columns parity tree (Proposition 8.9) dominates the two-phase tree at equal $O(n)$ cost. The constant is a coin-model number: under the pipeline's parity-locking the hard class vanishes and the unbounded floor on the literal pipeline is exactly 0 (Remark 8.11); the open remnant is the budgeted all-degrees floor.

7. **Adjacency certificates and the confirmation chain** (Section 9): two adjacent 1s certify a free pair *with certainty* (Lemma 9.1; a matched pair forces its row-neighbors to the determined 0), sound against the literal pipeline directly, with no design property used. The scan-then-confirm chain measures success 0.9317 at budget 1000 at both (32, 2) and (64, 4) (0.9417 on the exact pipeline at (32, 2)), posterior 1.0000 on every certified output, certificate rate $1 - e^{-B/400}$: the per-query caps are sharp, but they are per-query bounds, and the chain is the strongest known adaptive attacker on the pipeline, with the scaling of its constant the sharpest open problem (Open Problem 4).
8. **The two-phase certification tree, corrected** (Theorem 7.1): under the literal query semantics the tree’s error is exactly $2^{-(2d+1)} + (1 - 2^{-(2d+1)})2^{-2d+1}$ (the count-0 phase-2 branch); the earlier exact law $1 - (1 - f)2^{-(2d+1)}$ is retracted as a simulator artifact (Remark 7.2), the recorded 0.97-level measurements are requalified as hybrid-artifact measurements, and the literal tree measures 0.8500 at (32, 2) (Table 2); against the literal pipeline the tree is dead outright (Corollary 7.4).
9. **Budget reconciliation** (Section 10): Theorem 6.1 of [8] quantifies over $(d, d \log k)$ -trees; every constructed attacker is over budget, counting certificates are inert within budget, and the open problem is the budgeted certification floor, now organized by the certificate trichotomy (parity: dead; counting: inert within budget; adjacency: alive, Remark 9.4).
10. **Open problems** (Section 12): the chi-hypothesis in its printed budgeted form, the budgeted certification floor over the certificate trichotomy, the adjacency-budget race, chaining beyond the per-hit posterior, and the odd-characteristic analogue.

All numerical claims are exact channel simulations or exact enumerations over the finite coin model of Section 8 (they sample the proved channel law or enumerate its answer tables, not the design space), except where stated: the exact-pipeline rows of Section 9 sample uniform designs over $\text{Des}(4, 2)$ directly. The two-phase table separates two laws and labels every row: *hybrid-artifact* rows verify the artifact simulator’s certification law (Remark 7.2); *literal* rows verify Theorem 7.1. Against the literal pipeline the tree is dead outright: the row-sum channel answers the determined 0 on every pigeon (Corollary 7.4), a fact now proved at kernel level (Remark 3.5). The floor verification combines an exhaustive Bayes audit at (4, 1), exact enumeration of all 2^{20} coin matrices at (5, 2), and Monte Carlo at three points in one registered run. Measurements confirm the proved laws and are never used in place of proofs.

1.3 Attribution

The pipeline, the solution condition, the reduction chain, and the chi-task are due to J. Kraljček [8], with the algebraic foundation in [2]; we quote the printed definitions of [8] in Section 2. Everything else in this paper — the answer channel law, the degree-one span lemma, the completion lemma, the rigidity proposition, the factorization and depletion lemmas, Propositions 4.3, 4.4, 4.6 (in its surviving class form), Theorems 5.3, 6.1, 7.1 (in its corrected literal form), the two-phase tree itself, the floor lemmas and Theorem 8.7, the rows-and-columns parity tree (Proposition 8.9), and the adjacency certificate (Lemma 9.1) — is original to this research session and unpublished. This revision retracts two earlier claims of the session: the exact two-phase law $1 - (1 - f)2^{-(2d+1)}$ with its optimality conjecture (a simulator artifact, Remark 7.2) and the all-degrees non-adaptive cap (Remark 4.7). It also supersedes the previous draft’s design-space

readings discussion: the kernel-level verification (Remark 3.5) proves $V(n, d)^\rho = V(2d, d)$ exactly, so the outer/canonical debate dissolves, and the canonical-reading “reversed certificate” scenario of the previous draft does not exist under the zeroing restriction semantics (Remark 3.3, Proposition 3.4). We work from the definitions as printed; if the intended notion of the design set or of the tree class in [8] differs from our reading, our results apply to the printed formulation.

1.4 Notation

Symbol	Meaning
n	outer dimension: $n + 1$ pigeons, n holes; the outer system is $\neg\text{PHP}_n$
d	degree bound of the pipeline and of the queries; $d \geq 2$ is assumed where stated
p	field characteristic; throughout, $p = 2$ and the field is $\mathbb{F}_2$
x_{ij}	outer Boolean variable “pigeon i goes to hole j ”, $i \in [n + 1]$, $j \in [n]$
$\neg\text{PHP}_n$	system: hole collisions $x_{i_1 j} x_{i_2 j}$, two per pigeon $x_{i j_1} x_{i j_2}$, row sums $Q_i = 1 + \sum_j x_{ij}$, Boolean axioms $x_{ij}^2 + x_{ij}$
ρ	uniform random partial injection leaving exactly $2d$ holes unfilled
D, R	the $2d + 1$ free pigeons and the $2d$ free holes under ρ ; $F = D \times R$ the free pairs
L	uniform random degree- d design of the restricted system (Def. 2.1)
$V(N, d)$	span of the degree $\leq d$ consequences of a system on $N + 1$ pigeons, N holes
$\text{Des}(N, d)$	affine space of linear functionals with $L(1) = 1$ and $L _{V(N, d)} = 0$
$\omega = (\rho, L)$	pipeline sample; the answer to a query g is $L(g^\rho)$
g^ρ	restriction of g by ρ (killed variables become 0 or the constant 1)
f	free-pair density $= \mathbb{P}[(i, j) \in F] = \frac{(2d+1)2d}{(n+1)n}$
m	matched-pair density $= \mathbb{P}[\rho(i) = j] = \frac{n-2d}{(n+1)n}$
q	per-hit Bayes posterior $= \frac{f/2}{f/2+m} = \frac{2d^2+d}{2d^2-d+n}$
s, e	number of queries (budget); depth of a tree
e'	printed query budget of Theorem 2.3: $O(\log n) + O(d \log n)$, written $d \log k$
M	answer table $M(i, j) = \text{ans}(x_{ij})$ (Section 8)
err^*	the exact degree-1 certification floor (Theorem 8.7)
$\text{lab}(\ell)$	the pair output at leaf ℓ of a tree
$\chi(P, \rho)$	chi-indicator of [8]: 1 iff $\text{lab}(P) \notin F$ and $1 \in \text{Span}(W(P)^\rho)$
k, l	refutation size and AC depth in the target application; $d = (\log k)^{O(l)}$

All probabilities are over the uniform pipeline sample $\omega = (\rho, L)$: ρ is uniform over partial injections leaving exactly $2d$ holes free (the free hole set R , the free pigeon set D , and the bijection $[n + 1] \setminus D \rightarrow [n] \setminus R$ being chosen independently and uniformly), and L is uniform over $\text{Des}(2d, d)$, independently of ρ . Constants in $O(\cdot)$ and $o(\cdot)$ are absolute unless a parameter is displayed.

2 Preliminaries

2.1 The negated pigeonhole principle as a polynomial system

Work over $\mathbb{F}_2$. The outer system $\neg\text{PHP}_n$ on the variables x_{ij} , $i \in [n + 1]$, $j \in [n]$, consists of the hole-collision polynomials $x_{i_1 j} x_{i_2 j}$ ($i_1 < i_2$), the two-per-pigeon polynomials $x_{i j_1} x_{i j_2}$ ($j_1 < j_2$), the row sums $Q_i = 1 + \sum_{j=1}^n x_{ij}$ (in characteristic 2, minus is plus), and the Boolean axioms $x_{ij}^2 + x_{ij}$. The system expresses that the assignment is a total injection of $n + 1$ pigeons into n holes; it is inconsistent. For a rectangle of $N + 1$ pigeons and N holes let

$$V(N, d) = \text{Span}\{g \cdot h : h \text{ a generator, } \deg(gh) \leq d\} \subseteq \mathbb{F}_2^{\leq d}[x_{ij}]$$

N	d	$\dim \mathbb{F}_2^{\leq d}$	$\text{rank } V(N, d)$	$\dim \text{Des}(N, d)$
4	2	231	165	65
5	2	496	306	189
6	2	946	511	434
7	2	1653	792	860
8	2	2701	1161	1539
6	3	14190	12110	2079

Table 1: Exact $\mathbb{F}_2$ dimensions of the consequence space $V(N, d)$ inside the multiset-monomial space, and of the affine design space $\text{Des}(N, d)$, whose cardinality is $2^{\dim \text{Des}}$. In every case $1 \notin V(N, d)$ and the degree- $\leq d$ closure of V was verified by exact elimination.

be the span of its degree- $\leq d$ consequences, where $\mathbb{F}_2^{\leq d}$ carries the multiset monomials (x^2 is a distinct basis element from x). A *design* is a linear functional L on $\mathbb{F}_2^{\leq d}$ with $L(1) = 1$ and $L|_{V(N, d)} = 0$; designs exist precisely when $1 \notin V(N, d)$, which holds in the regime of interest (Razborov’s theorem as quoted in [8]; verified by exact $\mathbb{F}_2$ elimination in the cases of Table 1).

2.2 The systems ladder and where AC[2]-Frege sits

Resolution has exponential lower bounds for the pigeonhole principle [5]; so do $\text{Res}(k)$ and AC^0 -Frege [1]. The system $\text{AC}^0[p]$ -Frege (constant-depth Frege with MOD_p gates) is equivalent to the Extended Nullstellensatz algebraic proof system [7], and the circuit lower-bound technology for $\text{AC}^0[p]$ [11] has never lifted to it. In 2026 the adjacent frontier moved: unrestricted dag-like $\text{Res}(\oplus)$ was shown exponential for the bit pigeonhole principle [10], and random 3-CNFs got super-linear bounded-depth Frege bounds [9]. The AC[2]-Frege lower bound for $\neg\text{PHP}_n$ remains open, with no non-trivial bound known on any formula sequence; the pseudo-solution program of [8] is its most concrete current reduction.

2.3 The reduction chain of Krajíček

We quote the chain as printed in [8].

- **Theorem 2.2 of [8]** (from [2]): a k -step $\text{Fl}(\text{MOD}_p)$ refutation of $\neg\text{PHP}_n$ yields an Extended Nullstellensatz refutation with $S = k^{O(1)}$ extension polynomials in $l + O(1)$ levels, accuracy $h = O(\log S)$, and degree $d \leq (\log k)^{O(l)}$.
- **Theorems 3.2 and 3.3 of [8]**: if $e^{h/p} \geq 2S^2$ there is no $(d, h + \log S, S^{-1})$ -solution; contrapositively, a $((\log k)^{O(l)}, O(\log k), k^{-O(1)})$ -solution of $\neg\text{PHP}_n$ rules out k -step refutations.
- **Definition 3.1 of [8]** (the solution condition), quoted in Definition 2.2 below: every tree of the permitted class must fail, i.e. output a non-conflict, with probability at least $\gamma = k^{-O(1)}$.
- **Definition 4.3 of [8]** (the candidate $\Omega(n, d)$), quoted in Definition 2.1.
- **Theorem 6.1 of [8]** (the chi-task), quoted in Theorem 2.3.

Definition 2.1 (pipeline $\Omega(n, d)$; Definition 4.3 of [8]). Fix the characteristic p (here $p = 2$), outer dimension n , and degree d with $2d \leq n$. The pipeline $\Omega(n, d)$ is the distribution on pairs $\omega = (\rho, L)$:

1. ρ is a uniform random partial injection from $[n + 1]$ to $[n]$ leaving exactly $n_\rho = 2d$ holes unfilled; let $D \subseteq [n + 1]$ be the $2d + 1$ unassigned pigeons and $R \subseteq [n]$ the $2d$ unfilled holes. The restricted system is $\neg$ PHP on the rectangle $D \times R$, still contradictory ($2d + 1$ pigeons, $2d$ holes).
2. L is a uniform random degree- d design of the restricted system: a linear functional on the degree- $\leq d$ multiset monomials of the variables $\{x_{ij} : (i, j) \in D \times R\}$, vanishing on the restricted consequence space $V(n, d)^\rho$ (span of the restrictions of the outer degree- $\leq d$ consequences), with

$$L(1) = 1, \quad L|_{V(n, d)^\rho} = 0,$$

an element of a non-empty affine space (designs exist by the same separation as in Table 1; by Remark 3.5 the constraint space equals the canonical $V(2d, d)$, whose complementary dimensions the table records).

3. the answer to a query g , a polynomial of degree $\leq d$ in the outer variables, is $\omega(g) := L(g^\rho)$, where g^ρ replaces each outer variable x_{ij} by 1 if $\rho(i) = j$, by 0 if i is assigned and $j \neq \rho(i)$, and leaves it alone if $(i, j) \in D \times R$. (The printed clause leaves the cell *free pigeon, matched hole* — (i, j) with $i \in D, j \notin R$ — uncovered; the *zeroing completion* sets such cells to 0, the only choice consistent with the determined killed-unmatched answer, Remark 3.3.)

A pair (i, j) is *free* if $(i, j) \in F := D \times R$, *matched* if $\rho(i) = j$, and *killed-unmatched* otherwise.

One design space. An earlier draft discussed two “readings” of the design set — vanishing on $V(n, d)^\rho$ as printed (the *outer reading*) versus membership in the canonical $\text{Des}(2d, d)$ of the fresh system $\neg$ PHP on $D \times R$ — and treated their difference as material. A kernel-level verification (Remark 3.5) settles the question: $V(n, d)^\rho = V(2d, d)$ *exactly*, so the two readings coincide and the debate dissolves: there is only one design space, and the operative content of the printed vanishing condition is that it forces the free-row parities (Proposition 3.4) while leaving the free pairs fair on row-incomplete query sets (Theorem 3.2(iii)). All results below are stated for this single space.

Definition 2.2 (solution condition; Definition 3.1 of [8]). For parameters (d, e, γ) , a sample $\omega = (\rho, L)$ *satisfies the solution condition* if every (d, e) -tree T — an adaptive decision tree of depth e whose queries are polynomials of degree $\leq d$ and whose leaves are labelled by pairs — fails on ω with probability at least γ , where T *succeeds* on ω if $\text{lab}(T) \notin D \times R$ and $1 \in \text{Span}(W(T)^\rho)$, with $W(T)$ the span of the restricted system together with the query polynomials along the path of T . The family $\Omega(n, d)$ is a (d, e, γ) -pseudo-solution if the condition holds with probability 1.

Theorem 2.3 (chi-task; Theorem 6.1 of [8]). *If every $(d, d \log k)$ -tree T' satisfies*

$$\mathbb{P}_{\rho, P}[\chi(P, \rho) = 1] \geq k^{-O(1)},$$

where P is a random path of T' and $\chi(P, \rho) = 1$ iff $\text{lab}(P) \notin D^\rho \times R^\rho$ and $1 \in \text{Span}(W(P)^\rho)$, then every $\text{AC}^0[l](\text{MOD}_2)$ -Frege refutation of $\neg\text{PHP}_n$ has at least k steps.

The conjecture of [8] is that the hypothesis holds even with a constant lower bound at $k(n) = 2^{n^\delta}$ for small $\delta > 0$. The program can fail in two directions: if some tree succeeds with probability $> 1 - k^{-O(1)}$, the candidate $\Omega(n, d)$ is refuted as a pseudo-solution; if every tree's chi-probability is below $k^{-O(1)}$, Theorem 2.3 cannot be applied. The results of this paper bound the first quantity from above for large tree classes, thereby protecting the pseudo-solution, and isolate the exact residual open class.

3 The answer channel at characteristic 2

Throughout this section $(\rho, L) \sim \Omega(n, d)$ with $p = 2$ and $2d \leq n$; $d \geq 2$ is assumed where stated (the channel law (Theorem 3.2) holds for $d \geq 1$, the completion lemma needs $2d \geq 4$).

3.1 The degree-one span lemma

Recall the generators of the restricted system on $D \times R$: the collision polynomials $x_{i_1 j} x_{i_2 j}$, the two-per-pigeon polynomials $x_{ij_1} x_{ij_2}$, the row sums $Q_i^c = 1 + \sum_{j \in R} x_{ij}$, the Boolean axioms $x_{ij}^2 + x_{ij}$, and all their shifts by monomials of degree at most $d - \deg$. Write $\mathcal{G}$ for the set of their coefficient vectors in $\mathbb{F}_2^{\leq d}$, so that $V(2d, d) = \text{Span}(\mathcal{G})$, and recall the affine design space $\text{Des}(2d, d) = \{L : \langle L, g \rangle = 0 \ \forall g \in \mathcal{G}, \langle L, e_0 \rangle = 1\}$, where e_0 is the constant monomial and $\langle \cdot, \cdot \rangle$ is the coefficient pairing.

For a placement point z_{rv} ($r \in D, v \in R$) let $\text{eval}_{z_{rv}}$ be the linear functional “evaluate at the partial injection placing pigeon r on hole v and no other pigeon”, with x^2 evaluated as x .

Lemma 3.1 (degree-one span). *Assume $d \geq 2$ (so each row has $2d \geq 4$ cells). The degree- ≤ 1 part of $\text{Span}(\mathcal{G} \cup \{e_0\})$ is exactly $\text{Span}\{e_0, Q_1^c, \dots, Q_{2d+1}^c\}$: that is, a polynomial of degree ≤ 1 is a consequence of the restricted system (up to the constant) if and only if it is constant on each row.*

Proof. The inclusion $\supseteq$ is trivial: each Q_r^c is a generator and e_0 is present. For the inclusion $\subseteq$, let δ be a degree- ≤ 1 element of $\text{Span}(\mathcal{G} \cup \{e_0\})$ that is not constant on each row; we build a linear functional killing $\mathcal{G} \cup \{e_0\}$ but not δ , contradicting membership. Choose a row r^* on which δ is neither identically 0 nor identically 1, and let t be the number of cells of row r^* on which δ has coefficient 1; then $1 \leq t \leq 2d - 1$. Define a weight vector w on the cells of row r^* :

- if t is odd, let w be the all-ones vector of the row (weight $2d$, even);
- if t is even, let w be the indicator of one 1-cell of δ and one 0-cell of δ (both exist since $1 \leq t \leq 2d - 1$ and $2d - t \geq 1$).

In both cases w has even total weight, and the correlation $w \cdot \delta|_{r^*}$ equals $t \bmod 2 = 1$. Set

$$L^* = \sum_{v \in R} w_v \text{eval}_{z_{r^* v}}.$$

Every generator vanishes pointwise at every placement point (collisions and two-per-pigeon need two placed pigeons on a common hole or row; Booleans vanish on 0/1 points), hence their shifts do as well, so $\langle L^*, g \rangle = 0$ for every shifted collision, two-per-pigeon, and Boolean generator. For the row sums, $Q_u^c(z_{r^* v}) = [u \neq r^*]$, whence $\langle L^*, Q_u^c \rangle = [u \neq r^*] \cdot \sum_v w_v = [u \neq r^*] \cdot 0 = 0$, the total weight of w being even; likewise $\langle L^*, e_0 \rangle = 0$. Thus L^* annihilates $\mathcal{G} \cup \{e_0\}$, while

$\langle L^*, \delta \rangle = \sum_v w_v \delta_{r^*v} = 1$. A polynomial annihilated by no such functional cannot lie in the span; contradiction. Finally, the polynomials constant on each row are exactly the combinations $c e_0 + \sum_r a_r Q_r^c$ (row- r coefficient $c + a_r$), which proves the characterization.

For the pipeline's design space the constraint set is the span of the restrictions of the outer generators; by Remark 3.5 ($V(n, d)^\rho = V(2d, d)$ exactly) that span coincides with $\text{Span}(\mathcal{G})$, so the proof above covers the pipeline directly: the degree- ≤ 1 part of the constraint span (with e_0) is $\text{Span}\{e_0, Q_1^c, \dots, Q_{2d+1}^c\}$, and the row parities are forced to 1 on the free rows, the assigned rows carrying no free coordinates at all (their variables are zeroed by the restriction). $\square$

3.2 The channel law

Theorem 3.2 (answer channel law, single variables; the kernel law). *Let (i, j) be an outer pair and $a(x_{ij}) = L(x_{ij}^\rho)$ the answer to the single-variable query x_{ij} .*

- (i) *If $\rho(i) = j$ (matched), then $a = 1$ deterministically.*
- (ii) *If $i \notin D$ and $j \neq \rho(i)$ (killed-unmatched), then $a = 0$ deterministically.*
- (iii) *If $(i, j) \in F$ (free), then a is uniform on $\{0, 1\}$; moreover, on any row-incomplete query set S — no row of the rectangle has all $2d$ of its cells in S — the answers of the free pairs of S are jointly uniform: every pattern occurs for the same number of designs (Lemma 3.6(a)).*
- (iv) *Parity-locking. If the $2d$ cells of a free row are all queried, the answer pattern is locked to odd parity: exactly 2^{2d-1} of the 2^{2d} patterns occur, each for the same number of designs (Lemma 3.6(b)); the row XOR is 1 for every occurring pattern, and the all-zero pattern never occurs.*

Proof. (i) If $\rho(i) = j$ then $x_{ij}^\rho = 1$ and $a = L(1) = 1$ by the design axiom. (ii) If $i \notin D$ and $j \neq \rho(i)$ then $x_{ij}^\rho = 0$ and $a = 0$. (iii) Marginal uniformity: the coordinate $L \mapsto L(x_{ij})$ is constant on the affine design space if and only if the vector e_{ij} lies in the orthogonal complement of its direction space, which is $\text{Span}(\mathcal{G} \cup \{e_0\})$ (the restricted generators together with the constant). By Lemma 3.1 every degree- ≤ 1 element of that span is constant on each row; e_{ij} is supported on the single cell (i, j) and its row has $2d \geq 2$ cells, so it is not row-constant and not the constant vector. Hence the coordinate is non-constant, and a non-constant linear coordinate on an affine space over $\mathbb{F}_2$ takes the values 0 and 1 equally often: a is a fair bit. The joint statement is Lemma 3.6(a): the zeroing completion exhibits, for every row-incomplete pattern ψ on S , exactly $2^{\dim \text{Des}(2d, d) - |S|}$ designs, independently of ψ , so the answers of the free pairs of S are jointly uniform. (iv) If S contains a full free row r , Lemma 3.6(b) applies: extensions exist exactly for the pinned patterns of parity 1 on row r (rigidity (1)), and every admissible pattern occurs for the same number of designs, so the answer law is uniform on the 2^{2d-1} odd-parity patterns and gives nothing to the rest: the row is parity-locked, its XOR equal to 1. $\square$

Remark 3.3 (proof-route correction: no disjoint supports at $d \geq 2$). An earlier recorded route to the joint law (iii) argued from a kernel basis with pairwise-disjoint single-variable supports. That route fails at $d \geq 2$ and is superseded: the kernel's single-variable projection is the even-parity subspace of each free row, and within a row of $2d$ coordinates no $2d - 1$ even-weight vectors with pairwise disjoint supports exist, their total weight being at least $2(2d - 1) > 2d$ (the counting obstruction; at the restricted point $(4, 2)$ the reduced basis already has 15 overlapping single-variable pairs). At $(2, 1)$ the disjoint route happens to work; at every d the correct route is the

zeroing completion, Lemma 3.6, on which (iii) and (iv) rest. The completion itself is forced: the printed definition of g^ρ leaves the cell (free pigeon, matched hole) uncovered, and any completion leaving those variables symbolic would contradict the determined killed–unmatched answer (ii).

Proposition 3.4 (row-parity rigidity; the aggregate channel is closed). *Every design L of the restricted system satisfies*

$$\sum_{j \in R} L(x_{ij}) = 1 \quad \text{for every row } i \in D. \quad (1)$$

Consequently, in the pipeline, the outer row-sum query $Q_i = 1 + \sum_j x_{ij}$ answers the determined 0 on every pigeon: for a free pigeon i , $Q_i^\rho = Q_{\text{row}(i)}^c$ is itself a restricted consequence; for an assigned pigeon i with $\rho(i) = j_0$, the zeroing restriction gives $Q_i^\rho = 1 + x_{i,j_0}^\rho = 1 + 1 = 0$ identically. Row queries therefore carry no information at all, and no certificate built on Q_i exists against the pipeline. (An earlier draft derived, under a canonical reading, a “reversed certificate” with assigned pigeons answering 1; that computation treated assigned-row variables as symbolic, which the zeroing semantics excludes, and the scenario does not exist — the kernel-level verification confirms $L(Q_i^\rho) = 0$ under the single design space, Remark 3.5.)

Proof. The restricted row sum $Q_i^c = 1 + \sum_{j \in R} x_{ij}$ is a generator of degree $1 \leq d$, so $L(Q_i^c) = 0$; expanding with $L(1) = 1$ gives (1). For the outer query: if $i \in D$ then $Q_i^\rho = 1 + \sum_{j \in R} x_{ij} = Q_i^c$, whence $L(Q_i^\rho) = 0$. If $i \notin D$ with $\rho(i) = j_0$, the zeroing completion kills every cell of row i except the matched one, so $Q_i^\rho = 1 + 1 + \sum_j 0 = 0$ as a polynomial, whence $L(Q_i^\rho) = 0$. (The zeroing completion is the only one consistent with the determined killed–unmatched answer; Remark 3.3.) $\square$

Remark 3.5 (the one-kernel theorem). Exact $\mathbb{F}_2$ elimination (every generator row built through the outer route — outer polynomials, outer shifts, then restriction — and compared with the independently built canonical list by rank and by containment in both directions) gives $V(n, d)^\rho = V(2d, d)$ at every tested point, with the ranks matching the canonical values of Table 1:

outer (n, d)	restricted $(2d, d)$	rank $V(n, d)^\rho$	rank $V(2d, d)$	containment
(7,1)	(2,1)	3	3	both ways
(15,2)	(4,2)	165	165	both ways
(31,3)	(6,3)	12110	12110	both ways

The reason is structural: an assigned pigeon’s Q_i restricts to the zero polynomial ($1 + 1 = 0$), the killed Booleans, collisions, and two-hole monomials restrict to 0, and $\deg(h^\rho) = \deg(h)$ whenever $h^\rho \neq 0$, so every restricted outer generator is a canonical generator shift or 0. Three consequences are used below. First, the outer/canonical design-space debate of the previous draft dissolves: there is only one kernel, and the corpus’s adoption of the outer reading is moot as a definitional matter. Second, the printed vanishing condition itself forces the free-row parities (Proposition 3.4). Third, no design space satisfying the printed definition realizes the coin channel beyond its row-incomplete single-variable law: the “outer-design kernel” that masks the row parities in the never-queried killed coordinates violates the printed condition for about half its designs (measured 50382/100000 and 49962/100000), so the coin channel is exactly the row-incomplete single-variable answer law, the precise content of Corollary 3.8 with its E_{full} clause.

3.3 Completion of partial answer patterns

Lemma 3.6 (completion). *Assume $d \geq 2$. Let S be a set of cells of the rectangle $D \times R$ and $\psi : S \rightarrow \mathbb{F}_2$.*

- (a) *If no row contains all $2d$ of its cells in S , then ψ extends to a design: there is $L \in \text{Des}(2d, d)$ with $L|_S = \psi$; and the number of designs with $L|_S = \psi$ equals $2^{\dim \text{Des}(2d, d) - |S|}$, independently of ψ .*
- (b) *If S pins all cells of some row r , extensions exist only if the pinned values on row r have the parity forced by rigidity (parity 1 for a free row; assigned rows carry no free cells whose design values could be pinned). If every fully pinned row meets its parity requirement, extensions exist, and their number is $2^{\dim \text{Des}(2d, d) - |S| + \{\# \text{ fully pinned rows}\}}$, independently of the admissible ψ . In the applications below only free cells are ever pinned (killed cells answer determinedly and impose no design constraint), so every pinned row is a free row.*

Proof. The number of designs with $L|_S = \psi$ equals $2^{n - \text{rank}(A) - |S| + \rho_S}$, where n is the number of monomials, A the constraint system $(\mathcal{G}, e_0)$ of rank $\text{rank}(A)$, and ρ_S the dimension of the intersection of $\text{Span}\{e_u : u \in S\}$ with $\text{Span}(\mathcal{G} \cup \{e_0\})$; the solution set, when non-empty, is an affine space over the nullspace, giving the ψ -independence. So it suffices to compute ρ_S and feasibility.

(a) Suppose no row is fully pinned. If $\sum_u \delta_u e_u \in \text{Span}(\mathcal{G} \cup \{e_0\})$ for some δ supported on S , then by Lemma 3.1 the polynomial $\sum_u \delta_u x_u$ is constant on each row, say with value $c + a_r$ on row r ; but every row has a cell outside S , where the polynomial vanishes, forcing $c + a_r = 0$ for every r , i.e. the polynomial is identically zero and $\delta = 0$. Hence $\rho_S = 0$, the S -equations are independent of $(\mathcal{G}, e_0)$, every ψ is feasible, and the count is $2^{\dim \text{Des} - |S|}$, where $\dim \text{Des} = n - \text{rank}(A) - 1$.

(b) Suppose S pins the full rows of a set R^* of rows, and parity 1 holds on each of them. Any dependency now satisfies: on a non-fully-pinned row the polynomial vanishes off S , forcing $c + a_r = 0$ there; on a fully pinned row $r \in R^*$ the polynomial equals the constant $c + a_r$. Hence the dependency space is spanned by $\{\sum_{u \in \text{row}(r)} e_u : r \in R^*\}$, of dimension $|R^*|$, since $\sum_{u \in \text{row}(r)} x_u = Q_r^c + 1 \in \text{Span}(\mathcal{G} \cup \{e_0\})$ and no smaller combination lies in the span (the restriction of the dependency to a single fully pinned row is the all-ones pattern, which is not a dependence unless the whole row is used). Thus $\rho_S = |R^*|$ and the count is $2^{\dim \text{Des} - |S| + |R^*|}$ for every feasible ψ . Feasibility: rigidity (1) forces the parity condition, so it is necessary. For sufficiency, remove one pinned cell from each fully pinned row, obtaining S' with no full row; by (a) there is a design L' with $L'|_{S'} = \psi|_{S'}$, and by (1) the value of L' on each removed cell is forced to equal $1 - \sum_{\text{rest of its row}} \psi$, which is exactly the prescribed ψ -value by the parity hypothesis. Hence L' extends ψ . $\square$

3.4 The coin channel and its equivalence with the pipeline for single-variable trees

Definition 3.7 (coin channel). $\text{Coin}(n, d)$ is the channel with the same status process ρ as the pipeline, in which each free pair carries an independent fair bit, independent across pairs and of ρ , a matched pair answers 1 and a killed-unmatched pair answers 0, and the answer to a product $g = \prod x_{i_t j_t}$ is the $\mathbb{F}_2$ -product of the answers to its factors.

Corollary 3.8 (coin-channel equivalence for single-variable trees). *Let T be an adaptive tree of depth e all of whose queries are single outer variables, and let E_{full} be the event (under Ω or under Coin , identically) that the set of free pairs queried along the path of T contains all $2d$ cells of some row of the rectangle. Then*

$$\left| \mathbb{P}_{\Omega}[T \text{ outputs a free pair}] - \mathbb{P}_{\text{Coin}}[T \text{ outputs a free pair}] \right| \leq \mathbb{P}[E_{\text{full}}] \leq \min\left\{1, \frac{(n+1)\binom{e}{2d}}{\binom{n}{2d}}\right\}.$$

Proof. Conditionally on ρ and on the complement of E_{full} , Lemma 3.6(a) gives that the probability, over the design space, of any fixed answer pattern on the queried free pairs equals $2^{-a(\rho)}$, where $a(\rho)$ is the number of queried free pairs, independently of the pattern; the killed pairs answer determinedly in both models. This is exactly the coin-channel likelihood. Equal likelihoods given ρ imply equal joint laws of $(\rho, \text{transcript})$ off E_{full} , hence equal output laws off that event; on the event the two laws may differ, and the total variation between the output distributions is at most the probability of the event. For the bound: if the path queries s_r cells of row r , then the row's free pairs are all queried only if $s_r \geq 2d$ and the row's free set is a $2d$ -subset of the queried cells; by symmetry of ρ , a fixed $2d$ -set is the free set of a fixed row with probability $\binom{n}{2d}^{-1}$, and a union bound over rows and over the at most $\binom{e}{2d}$ subsets of the queried cells gives the claim. $\square$

Thus for the single-variable class the coin channel is not an idealization but an exact law up to the negligible event E_{full} ; in the parameter regime of interest, $e = d \log k \ll n$, the bound is $o(1)$ (and at $(n, d, e) = (32, 2, 33)$ it is below 4×10^{-3}). The kernel-level verification sharpens the statement to its exact form: off E_{full} the total variation between the two answer laws is exactly 0 (on every marginal and on every row-incomplete joint, confirmed at the points of Remark 3.5); on a fully queried free row it is exactly $\frac{1}{2}$, the design law being uniform on the 2^{2d-1} odd patterns against the coin law's 2^{2d} ; and the coin channel is realized by no design space satisfying the printed definition (Remark 3.5). The coin-channel samplers accordingly over-sample: a coin sample satisfies the design constraints with probability $\approx 2^{-(2d+1)}$ (measured 0.12587, 0.03051, 0.00793 against $2^{-3}, 2^{-5}, 2^{-7}$ at the restricted points $(2, 1), (4, 2), (6, 3)$).

4 Fixed labels and non-adaptive trees

Throughout, $f = \mathbb{P}[(i, j) \in F] = \frac{(2d+1)2d}{(n+1)n}$ and $m = \mathbb{P}[\rho(i) = j] = \frac{n-2d}{(n+1)n}$; note $f = O(d^2/n^2)$ and $f + m \leq 2/n$ whenever $2d \leq \sqrt{n}$.

The following trivial but fundamental disintegration underlies all caps. Recall that a *status pattern* σ assigns to each cell of a set S one of the labels free, matched, killed–unmatched.

Lemma 4.1 (factorization). *Let A be any event determined by the answers to a fixed set of queries, and let $\sigma(A)$ denote the σ -algebra generated by the status patterns of the queried cells that are consistent with those answers. For every event B depending only on ρ (such as $p \in F$):*

$$\mathbb{P}[B \mid A] = \sum_{\sigma} \mathbb{P}[\sigma \mid A] \mathbb{P}[B \mid \sigma].$$

Proof. Given a status pattern σ , the answers are a deterministic function of the status pattern and the design values of the free queried coordinates; the latter are, conditionally on σ , independent of ρ (the design is sampled independently of ρ , and σ is ρ -measurable). Hence

$\mathbb{P}[B \mid \sigma, \text{answers}] = \mathbb{P}[B \mid \sigma]$ for every ρ -event B , and the tower rule applied to the partition generated by $(\sigma, \text{answers})$ gives the claim. $\square$

Lemma 4.2 (hypergeometric depletion). *Let S be a set of at most s cells and let σ be a status assignment on S . Then for every cell $p \notin S$,*

$$\mathbb{P}[p \in F \mid \text{status}(S) = \sigma] \leq \frac{(2d+1)}{(n+1-s)} \cdot \frac{2d}{(n-s)} = f \left(1 + O\left(\frac{s}{n}\right)\right) \quad (s \leq n/2),$$

and the same bound holds conditionally on any event determined by the answers, by Lemma 4.1. The pigeon-freeness indicators $\{[i \in D]\}_i$ and the hole-freeness indicators $\{[j \in R]\}_j$ are inclusion indicators of without-replacement samples and are negatively associated in the sense of Joag-Dev and Proschan [6] (their Section 3 and Theorem 8); the pair indicators $[p \in F] = [i \in D] \cdot [j \in R]$ inherit the negative association by the closure of negative association under products over disjoint supports, in the occupancy formulation of Dubhashi and Ranjan [4].

Proof. Reveal the statuses on S ; let r_p be the number of pigeons other than p 's pigeon revealed free, and r_h the number of holes revealed free. The completions of σ to full partial injections form a without-replacement scheme: the free pigeons of the completion are a uniform subset of the right size, and similarly for holes. Direct counting gives $\mathbb{P}[i \in D \mid \sigma] = \frac{2d+1-r_p}{n+1-s'}$ and $\mathbb{P}[j \in R \mid \sigma] = \frac{2d-r_h}{n-s'}$ with $s' \leq s$ the number of cells of S touching the relevant coordinate, and the joint probability satisfies

$$\mathbb{P}[p \in F \mid \sigma] \leq \frac{2d+1}{n+1-s} \cdot \frac{2d}{n-s} = \frac{(2d+1)2d}{(n+1)n} \cdot \frac{(n+1)n}{(n+1-s)(n-s)} = f(1 + O(s/n)),$$

the joint inequality being the negative dependence of sampling without replacement: the freeness indicators of the pigeons are the inclusion indicators of a uniform without-replacement sample, a canonical negatively associated family [6], and the product form for $p \in F$ follows from the closure under products of negatively associated families over disjoint index sets [6], in the occupancy formulation of [4]. The final sentence is Lemma 4.1 with $B = \{p \in F\}$. $\square$

Proposition 4.3 (fixed labels). *Let T be any tree, with queries of any degree and any depth, all of whose leaves carry the same label (i_0, j_0) . Then its success probability (output a free pair on a live path) equals exactly*

$$\mathbb{P}[(i_0, j_0) \in F] = \frac{(2d+1)}{(n+1)} \cdot \frac{2d}{n} = f,$$

independent of the queries, the depth, and the answers.

Proof. Success is the event that the fixed output pair is free, which depends on ρ alone. The pipeline is invariant under the full pigeon and hole permutation groups (the design space $\text{Des}(2d, d)$ is invariant, L being uniform over it), so every one of the $(n+1)n$ pairs is free with the same probability; the expected number of free pairs is $|D||R| = (2d+1)2d$, so the common value is $(2d+1)2d/((n+1)n) = f$. The queries are irrelevant because the label is fixed. $\square$

Proposition 4.4 (non-adaptive single-variable trees with answer-seeking labels). *Let T be a non-adaptive tree with single-variable queries $q_1, \dots, q_s$ fixed in advance and the leaf label a*

function of the s answers. Assume the label is either a pair that answered 1, or, if all answers are 0, an unqueried pair. Then

$$\mathbb{P}[T \text{ succeeds}] \leq \frac{sf}{2} + f\left(1 + O\left(\frac{s}{n}\right)\right) + o(1) = f\left(1 + \frac{s}{2}\right) + O\left(\frac{sf}{n}\right) + o(1),$$

and the error is at least $\frac{1}{2} - o(1)$ whenever $s \leq n/2$ and $2d \leq \sqrt{n/2}$.

Proof. Let A_1 be the event that some answer equals 1. On A_1 the label is a pair that answered 1; success requires some queried pair to answer 1 while free, and the label can coincide with at most one of them, so by the union bound and linearity (no independence needed; each q_t is a fixed pair)

$$\mathbb{P}[\text{success}, A_1] \leq \sum_{t=1}^s \mathbb{P}[q_t \in F \text{ and } a(q_t) = 1] = \sum_{t=1}^s \mathbb{P}[q_t \in F] \cdot \frac{1}{2} = \frac{sf}{2},$$

using Theorem 3.2(iii) and symmetry. On the complement A_0 (all answers 0) the label is an unqueried pair p determined by the all-zero pattern, and Lemma 4.2 applied to the s queried cells gives $\mathbb{P}[p \in F \mid A_0] \leq f(1 + O(s/n))$. Summing gives the bound. The numeric claim: $f(1 + s/2 + O(s/n)) \leq \frac{4d^2}{n^2}(1 + \frac{n}{4} + O(1)) \leq \frac{d^2}{n} + o(1) \leq \frac{1}{2}$ for $2d \leq \sqrt{n/2}$ and $s \leq n/2$. $\square$

Remark 4.5. Without the fallback discipline (a tree may output a queried pair that answered 0, which is free with probability $f/2$ per query), the $sf/2$ term doubles to sf ; that case is covered by Proposition 4.6 within the variable/monomial class. The measured adaptive first-hit success 0.2367 at $(n, d, s) = (32, 2, 33)$ sits below the bound $f(1 + s/2) = 0.256$, as it must.

Proposition 4.6 (non-adaptive variable/monomial-class trees). *Let T be a non-adaptive tree with $s \leq n/2$ fixed queries, each of which is either a single variable or a monomial in at most γ distinct variable pairs (so $\gamma = 1$ for single-variable queries and $\gamma = 2$ for degree-2 monomials), the label an arbitrary function of the answers, possibly an unqueried pair. Then*

$$\mathbb{P}[T \text{ succeeds}] \leq (\gamma s + 1) f\left(1 + O\left(\frac{s}{n}\right)\right) + o(1).$$

Consequently, at the program's budget $s = e' = d \log k$, the chi-task error requirement $k^{-O(1)}$ is safe within this class as long as $(\gamma s + 1)f \leq 1 - k^{-O(1)}$, i.e. up to $d = \Theta_\gamma(n^{2/3}(\log k)^{-1/3})$; beyond that the bound ceases to decide the hypothesis.

Proof. Let Q be the set of variable pairs occurring in the queries, $|Q| \leq \gamma s$. Split by whether the output is in Q :

$$\mathbb{P}[\text{success}] = \mathbb{P}[\text{success}, \text{lab} \in Q] + \mathbb{P}[\text{success}, \text{lab} \notin Q].$$

Queried output: for each $q \in Q$, $\mathbb{P}[\text{lab} = q, q \in F] \leq \mathbb{P}[q \in F] = f$ by symmetry (Proposition 4.3); the union bound gives the first term at most $|Q|f \leq \gamma sf$.

Unqueried output: condition on the full answer pattern π ; the label is a fixed cell $p(\pi) \notin Q$. The answers reveal information about ρ only through the status pattern of the queried cells (Lemma 4.1: for variables and for monomials under the product semantics the answer of each query is a deterministic function of the answers to its at most γ component pairs; on the literal pipeline the monomial answer is instead a fresh value of the same data, and Lemma 4.1 needs only that each answer be a function of the queried cells' statuses and design values, the latter

independent of ρ), and by Lemma 4.2 every status assignment σ satisfies $\mathbb{P}[p(\pi) \in F \mid \sigma] \leq f(1 + O(s/n))$; averaging over σ with the weights $\mathbb{P}[\sigma \mid \pi]$ gives $\mathbb{P}[p(\pi) \in F \mid \pi] \leq f(1 + O(s/n))$, and averaging over π gives the second term.

At $s = d \log k$, $(\gamma s + 1)f = \Theta(\gamma d^3 \log k / n^2)$, which is at most $1 - k^{-O(1)}$ exactly up to $d = \Theta_\gamma(n^{2/3}(\log k)^{-1/3})$. $\square$

Remark 4.7 (retraction of the all-queries generality). Proposition 4.6 was previously claimed for arbitrary queries of any degree, with the bound $(s + 1)f + o(1)$ and the transition $d > \Theta(n^{2/3}(\log k)^{-1/3})$ read as an all-degrees statement. That generality is **false** and is retracted. The rc counting tree — non-adaptive, $s = 2n + 1$: query all $n + 1$ row sums Q_i and all n column sums $K_j = 1 + \sum_i x_{ij}$, certify every line whose count of 1s differs from 1, and output a certified row's pigeon paired with a certified column's hole — achieves exact success $1 - 3 \cdot 2^{-(2d+1)}$ at $s = 2n + 1$ queries: 0.9067 at $(64, 2)$, against the retracted cap's $(s + 1)f = 130f = 0.625$ at the same point. The retracted proof's per-query accounting implicitly required the answer-1 event of a query to carry $O(f)$ of free mass; that holds for single variables and bounded-degree monomials, whose answer-1 event entails all γ component pairs to be free or matched, but fails for linear forms: the row sum touches the $2d$ free cells of its row and answers 1 with probability $\approx \frac{1}{2} \gg f$. The counting certificates that break the general claim are exactly the line-count certificates of Lemma 8.4 (Section 8), where they are proved sound and shown to be optimal.

The argument uses only pair symmetry, the determined-answer law, and depletion, so Proposition 4.6 holds at every prime p and under either semantics for product queries, within the variable/monomial class.

5 Adaptive single-variable trees

Lemma 5.1 (own-answer Bayes bound). *In the coin channel, let p be a pair queried $h \geq 1$ times along a path (necessarily with the identical answer repeated, since the same design bit is re-read). Then*

$$\mathbb{P}[p \in F \mid a_1, \dots, a_h] \leq \max(q, f), \quad q = \frac{f/2}{f/2 + m} = \frac{2d^2 + d}{2d^2 - d + n},$$

with equality throughout for a single answer 1. The same bound holds in the pipeline up to the event E_{full} (Corollary 3.8).

Proof. Given only the pair's own answers, its status is matched (all answers 1, likelihood 1), free (the single design bit, likelihood $\frac{1}{2}$ for any consistent pattern), or killed-unmatched (all answers 0, likelihood 1); the priors are $m, f, 1 - f - m$. Bayes gives

$$\mathbb{P}[p \in F \mid a_1 = \dots = a_h = 1] = \frac{f/2}{f/2 + m} = q, \quad \mathbb{P}[p \in F \mid \text{some } a_t = 0] = \frac{f/2}{f/2 + 1 - f - m} \leq f.$$

The transfer to the pipeline is Corollary 3.8. $\square$

Lemma 5.2 (one-cell perturbation). *In the coin channel, fix a cell p and a partial answer sequence T' for the other queried cells ($|T'| \leq s$). Then for every pair of statuses $s_1, s_2 \in \{F, M, U\}$ of p ,*

$$\frac{\mathbb{P}[T' \mid \sigma_p = s_1]}{\mathbb{P}[T' \mid \sigma_p = s_2]} \in [1 - C_n^s, 1 + C_n^s] \quad \text{whenever } s \leq n/2,$$

for an absolute constant C , where σ_p denotes the status of p . The same holds in the pipeline off the event E_{full} of Corollary 3.8.

Proof. Reveal T' answer by answer, and condition throughout on the revealed prefix and on σ_p . Conditional on the prefix, the next answer's law is determined by the conditional composition of the pools: the probability that the next queried cell is matched, free, or killed-unmatched is a hypergeometric quantity in the remaining pigeon and hole pools. The two hypotheses $\sigma_p = s_1$ and $\sigma_p = s_2$ differ by the position of one fixed pigeon-hole pair (free in one, matched or absent in the other), and by the elementary finite-population count of Lemma 4.2 every conditional status probability of an as-yet-unqueried cell differs between the two branches by a factor $1 + O(1/n)$ (one consumed or freed pair shifts the numerators and denominators of the same hypergeometric ratios by at most one). Conditional on the prefix and the status of the next cell, its answer is either determined identically in both branches, or it is a fresh fair design bit in both branches; hence each per-step likelihood ratio lies in $[1 - C/n, 1 + C/n]$, and after at most s steps the ratio lies in $[1 - Cs/n, 1 + Cs/n]$. The pipeline statement is Corollary 3.8: off E_{full} the two channels have equal likelihoods for single-variable transcripts. $\square$

Theorem 5.3 (adaptive single-variable cap). *Let T be an adaptive tree of depth e all of whose queries are single outer variables. Then, in the pipeline at $p = 2$,*

$$\mathbb{P}[T \text{ succeeds}] \leq \max(q, f) \left(1 + O\left(\frac{e}{n}\right)\right) + o(1), \quad q = \frac{f/2}{f/2 + m} = \frac{2d^2 + d}{2d^2 - d + n} \approx \frac{2d^2}{2d^2 + n}.$$

In particular, for $2d \leq \sqrt{2n}$ we have $q \leq \frac{1}{2} + o(1)$ and $f = o(1)$, so every such tree fails with probability at least $\frac{1}{2} - o(1) \geq k^{-O(1)}$ for every polynomial k : the chi-task hypothesis of Theorem 2.3 holds for the entire adaptive single-variable class at $p = 2$ in the super-polynomial regime.

Proof. By Corollary 3.8 it suffices to prove the bound in the coin channel and add the E_{full} slack, which is absorbed into $o(1)$. Decompose success over the leaves ℓ :

$$\mathbb{P}[\text{success}] = \sum_{\ell} \mathbb{P}[\text{reach}(\ell)] \mathbb{P}[\text{lab}(\ell) \in F \mid \text{reach}(\ell)],$$

the reach events being disjoint. Fix a leaf with transcript T_{ℓ} (queries and answers along the path) and label p .

Case 1: p occurs among the path's queries. Let $a(p)$ be its own answers and T' the remaining answers. Condition on the status σ_p of p ; by Lemma 4.1 (the design values carry no information about ρ beyond the statuses), the posterior odds between any two statuses of p given the full transcript equal the prior odds times the own-answer likelihood ratio times the perturbation factor of Lemma 5.2:

$$\frac{\mathbb{P}[\sigma_p = s_1 \mid T_{\ell}]}{\mathbb{P}[\sigma_p = s_2 \mid T_{\ell}]} = \frac{\mathbb{P}[\sigma_p = s_1]}{\mathbb{P}[\sigma_p = s_2]} \cdot \frac{\mathbb{P}[a(p) \mid s_1]}{\mathbb{P}[a(p) \mid s_2]} \cdot \Theta, \quad \Theta \in (1 - O(\frac{e}{n}), 1 + O(\frac{e}{n})),$$

with priors $f, m, 1 - f - m$ for F, M, U . If $a(p)$ is all-ones, $\mathbb{P}[a(p) \mid F] = \frac{1}{2}$ and $\mathbb{P}[a(p) \mid M] = 1$, while $\mathbb{P}[a(p) \mid U] = 0$: hence

$$\mathbb{P}[p \in F \mid T_{\ell}] = \frac{(f/2) \Theta_F}{(f/2) \Theta_F + m \Theta_M} \leq \frac{f/2}{f/2 + m} (1 + O(e/n)) = q(1 + O(e/n)),$$

using $\Theta_F/\Theta_M \in (1 - O(e/n), 1 + O(e/n))$. If $a(p)$ contains a 0, then $\mathbb{P}[a(p) \mid M] = 0$ and $\mathbb{P}[a(p) \mid U] = 1$, so the posterior is at most $\frac{f/2}{f/2+(1-f-m)}(1 + O(e/n)) \leq f(1 + O(e/n))$. In both sub-cases $\mathbb{P}[p \in F \mid T_\ell] \leq \max(q, f)(1 + O(e/n))$, which is Lemma 5.1 relaxed by the perturbation factor.

Case 2: p is not queried on the path. Then T_ℓ reveals no own-answer for p , and Lemma 4.2 with S the queried cells gives $\mathbb{P}[p \in F \mid T_\ell] \leq f(1 + O(e/n))$ directly.

Summing over leaves and using $\sum_\ell \mathbb{P}[\text{reach}(\ell)] \leq 1$ completes the proof. For the corollary, $q \leq \frac{1}{2} + o(1)$ is equivalent to $4d^2 \leq (1 + o(1))n$, i.e. $2d \leq (1 + o(1))\sqrt{2n}$. $\square$

Remark 5.4 (scope: Theorem 5.3 is a budgeted cap). The coupling slack $O(e/n)$ is budget-dependent, and the bound is vacuous at $e \asymp n^2$. The full-scan counting tree of Theorem 8.7 queries $n(n+1)$ single variables and succeeds with probability $1 - 2^{-\Theta(d^2)}$; this does *not* contradict Theorem 5.3: the counting tree lives where the slack has exploded. At the program’s budget $e = d \log k \ll n$ the slack is $o(1)$ and Theorem 5.3 is the strongest proved barrier for the adaptive single-variable class. Reconciling the unbounded and budgeted quantifiers is Section 10.

Remark 5.5 (measured confirmations). Exact channel simulations (sampling the proved channel law) confirm Theorem 5.3 and Lemma 5.1.

1. At $(n, d) = (32, 2)$: the analytic posterior is $q = \frac{2d^2+d}{2d^2-d+n} = \frac{10}{38} = 0.2632$; the measured freeness posterior given answer 1 was 0.2367 over 300 simulations, against the base rate $f = 0.0189$, and the first-hit scan over 33 queries succeeded 0.2367.
2. Threshold scan at $n = 128$: first-hit success tracks the posterior across $d = 2, 4, 8, 11, 16, 24, 32$: measured success 0.018, 0.072, 0.246, 0.400, 0.656, 0.898, 0.970, against the posterior $q = 0.075, 0.231, 0.548, 0.705, 0.846, 0.936, 0.970$; the posterior crosses $\frac{1}{2}$ at $d \approx \sqrt{n/2} = 8$ as predicted, and the scan’s own success crosses shortly above it, the offset being the budget factor $1 - (1 - p_1)^s$ with per-query hit probability $p_1 = f/2 + m$.
3. Scaling at fixed $(d, s) = (2, 25)$: success 0.1450, 0.0425, 0.0125 at $n = 32, 64, 128$ (400 simulations each), an empirical exponent of -1.77 against the cap model’s -2 : the freeness signal decays polynomially in n exactly as the rare-event analysis predicts, while $k^{-O(1)} = 2^{-n^{\delta O(1)}}$ decays exponentially, so the hypothesis’ margin grows rapidly in the conjecture’s regime.
4. Adaptive degree-2 strategies at $(d, s) = (2, 40)$: single-variable 0.183, 0.067, 0.013; hybrid (single plus AND) 0.250, 0.083, 0.020; pure AND-scan 0.027, 0.003, below 10^{-3} ; at $n = 32, 64, 128$: empirical exponents -1.8 to -1.9 ; no tested adaptive degree-2 strategy decays more slowly than n^{-2} (see Section 6 and Open Problem 2). (The tested “hybrid” branch was later found to have degenerated to the plain scan — its product test was never implemented — so this probe never exercised adjacency-conditioned certification; the dedicated chaining study of Section 9 supplies the missing test and reverses the empirical conclusion at the tree level.)

6 Degree-2 product queries: the no-lift theorem

At $p = 2$ we formalize the answer to a product query $g = x_{ab} \cdot x_{cd}$ by the multiplicative evaluation semantics, as in the session’s simulation harness and consistently with the channel law:

$$a(g) = a(x_{ab}) \cdot a(x_{cd}),$$

so $a(g) = 1$ iff both factors evaluate to 1. (See Remark 6.2 for the strictly linear reading $L(g^\rho)$; the two agree except when both component pairs are free.)

Theorem 6.1 (AND-query no-lift). *In the pipeline at $p = 2$ (equivalently, in the coin channel), let $g = x_{ab} \cdot x_{cd}$ with distinct component pairs and let $pa = (a, b)$ be the first component. Then*

$$\mathbb{P}[a(g) = 1] = \frac{f^2}{4} + fm + m^2, \quad \mathbb{P}[a(g) = 1 \text{ and } pa \in F] = \frac{f^2}{4} + \frac{fm}{2},$$

and consequently the per-hit labeling posterior is

$$\mathbb{P}[pa \in F \mid a(g) = 1] = \frac{f^2/4 + fm/2}{f^2/4 + fm + m^2} = \frac{f}{f + 2m} = \frac{f/2}{f/2 + m} = q,$$

exactly the single-variable posterior: the AND query is a rarer joint test whose per-hit posterior coincides with the single-variable Bayes posterior.

Proof. Enumerate the joint statuses of the two component pairs, using that the answers of distinct free pairs are independent fair bits (Theorem 3.2(iii) with Lemma 3.6(a)) and that matched and killed–unmatched answers are determined:

statuses of (pa, pb)	probability	$\mathbb{P}[a(g) = 1]$	$\mathbb{P}[a(g) = 1, pa \in F]$
both free	f^2	1/4	1/4
pa free, pb matched	fm	1/2	1/2
pa matched, pb free	fm	1/2	0
both matched	m^2	1	0
some killed–unmatched	$1 - (f + m)^2$	0	0

For “both free” the answer is the product of two independent fair design bits, equal to 1 with probability $\frac{1}{4}$, and pa is then free with certainty. For “free–matched” the matched factor contributes its determined 1 and the free factor its fair bit. Column sums give the two displayed identities. For the posterior, multiply numerator and denominator by $4/f$ (a polynomial identity in (f, m) , hence exact):

$$\frac{f^2/4 + fm/2}{f^2/4 + fm + m^2} = \frac{f + 2m}{f + 4m + 4m^2/f} = \frac{f(f + 2m)}{f^2 + 4fm + 4m^2} = \frac{f}{f + 2m} = \frac{f/2}{f/2 + m},$$

using $(f + 2m)^2 = f^2 + 4fm + 4m^2$. □

The matched–matched mass dominates $\mathbb{P}[a(g) = 1]$ and contributes zero labeling success, exactly canceling the both-free advantage: within the class of trees whose degree-2 tests are single-pass, no lift over the single-variable posterior is possible. At the tree level the picture changed with the adjacency mechanism (Section 9): the lift exists, through certainty events rather than better posteriors, and it needs single variables, not products. For degree-2 chains specifically, budget-honest accounting shows no lift: the AND-scan is *worse* than the single-variable scan (0.1333 against 0.2550 at $(32, 2)$, budget 1000), and AND-split chains that certify through adjacency fire far more rarely than the degree-1 chain (Open Problem 2).

Remark 6.2 (the strictly linear reading). If instead the answer to $g = x_{ab}x_{cd}$ is the linear value $L(g^\rho)$, then when both component pairs are free the answer is the design value of the degree-2 monomial, itself a fair bit provided that monomial is not a consequence of the restricted system (generic for distinct rows and columns). The table then changes only in the “both free” row,

from $\frac{1}{4}$ to $\frac{1}{2}$, and the posterior becomes $\frac{f(f+m)}{f^2+2fm+2m^2}$, equal to 0.311 at $(32, 2)$ against 0.263; simultaneously the hit probability falls by the loss of the matched-matched mass from the both-free row's complement, i.e. the test becomes rarer per unit of posterior gain, and the no-lift conclusion for first-hit strategies at fixed budget is unaffected in order of magnitude. Open Problem 2 subsumes both readings.

Remark 6.3 (pipeline degree-2 laws; transfer status of Theorem 6.1). On the exact pipeline the product semantics of this section is not the design law: the same-hole and same-pigeon degree-2 monomials are consequences (determined 0, where the product semantics answers 1 with probability $\frac{1}{4}$), the diagonal monomials are fresh fair bits independent of their two component singles (the product semantics correlates them perfectly), and the square law $L(x^2) = L(x)$ is the one identity the product semantics gets right. The per-hit no-lift of Theorem 6.1 is derived under the product semantics; its pipeline transfer is not covered by the proof but is measured-consistent: true monomial queries on the exact pipeline land on the cap, 0.2629 [0.2129, 0.3199] against $q = 0.2632$ at $(32, 2)$, budget 1000 (the `mono2` row of the chaining study). The transfer is listed as coin-model-only in Table 3.

7 The two-phase certification tree

7.1 The tree and its exact error law in the coin channel

The two-phase certification tree $\mathcal{T}_{2\text{ph}}$ is the adaptive degree-1 tree:

Phase 1. For pigeons $i = 1, \dots, n+1$: query the row sum $Q_i = 1 + \sum_j x_{ij}$; stop at the first answer 1 and set i^* to that pigeon. (In the coin channel an assigned pigeon answers the determined 0; a free pigeon answers 1 plus the XOR of its $2d$ free-row design bits, a fair coin: answer 1 iff the row's free count is *even*.)

Phase 2. For holes j : query x_{i^*j} ; stop at the first answer 1 and label the leaf (i^*, j) .

Fallback. If all queries of a phase answer 0, label an arbitrary pair.

Theorem 7.1 (literal error law of the two-phase tree, coin channel). *In $\text{Coin}(n, d)$, under the literal query semantics (the answer to Q_i is 1 iff the row XOR is 0), the two-phase tree errs with probability exactly*

$$\text{err}(\mathcal{T}_{2\text{ph}}) = 2^{-(2d+1)} + (1 - 2^{-(2d+1)}) 2^{-2d+1} = 2^{-2d} \left(\frac{5}{2} - 2^{-2d} \right) = 5 \cdot 2^{-(2d+1)} (1 + o(1)),$$

up to the fallback's base-rate luck: with the fallback pair free with probability f , the total error is $(1 - f) \text{err}(\mathcal{T}_{2\text{ph}})$. The first term is the phase-1 miss (every free pigeon's row answers even); the second is the count-0 branch: the certified pigeon's row has an even count of 1s, and all-zero is even, an event of conditional probability $2^{-2d}/\frac{1}{2} = 2^{-2d+1}$.

Proof. Phase 1 queries the row sums in order. An assigned pigeon answers the determined 0: its row XOR is 1 (the matched cell), so $Q_i^0 = 1 + 1 = 0$. A free pigeon's answer is 1 plus the XOR of its $2d$ free-row design bits: an independent fair coin, the events across the $2d + 1$ free pigeons being independent (disjoint rows, independent coordinates). An answer 1 certifies freeness soundly (assigned pigeons never produce it), and phase 1 misses with probability exactly $2^{-(2d+1)}$.

(n, d)	law	measured	predicted	simulations
<i>hybrid-artifact rows: certify on “free and row-XOR odd” (Remark 7.2)</i>				
$(32, 2)$	$1 - (1 - f)2^{-5}$	0.970	0.969	500
$(64, 2)$	$1 - (1 - f)2^{-5}$	0.97–0.98	0.969	500
$(128, 2)$	$1 - (1 - f)2^{-5}$	0.974	0.969	500
$(64, 4)$	$1 - (1 - f)2^{-9}$	0.998	0.998	500
<i>literal-tree rows: the law of Theorem 7.1</i>				
$(32, 2)$	$1 - (1 - f) \left[2^{-5} + (1 - 2^{-5})2^{-3} \right]$	0.8500	0.8505	2000
$(64, 2)$	$1 - (1 - f) \left[2^{-5} + (1 - 2^{-5})2^{-3} \right]$	0.8380	0.8484	1500
$(64, 4)$	$1 - (1 - f) \left[2^{-9} + (1 - 2^{-9})2^{-7} \right]$	0.9888	0.9904	800
$(32, 2)$	retry: $1 - (1 - f)(9/16)^5$	0.9430	0.9448	2000

Table 2: Measured success of the two-phase certification tree, separated by law. Hybrid rows: the artifact simulator of Remark 7.2, whose certification condition no degree-1 query implements; they verify the hybrid law, not the tree. Literal rows: an exact literal-law re-implementation confirming Theorem 7.1. Exact enumeration at $(5, 2)$ (all 2^{20} coin matrices) gives strict values $0.8476562 = (31/32)(7/8)$ for the literal tree and $0.9436865 = 1 - (9/16)^5$ for the literal retry, both digit-exact; the published-law tree re-measures 0.9695 at $(32, 2)$, reproducing the corpus’s 0.97 (flipped-law audit only).

Given certification, the certified pigeon’s row has an even count. Phase 2 scans that row (killed holes answer the determined 0) and succeeds iff the count is at least 1; conditionally on evenness the row’s answer pattern is uniform over the 2^{2d-1} even patterns, of which exactly one is all-zero, so the failure probability is $2^{-2d}/\frac{1}{2} = 2^{-2d+1}$. The earlier draft asserted this step deterministic (“phase 2 never fails given certification”); that assertion is precisely the gap corrected here: an even row count need not contain a 1.

Total, with the fallback firing on either failure and its label free with the base rate f :

$$\text{err}(\mathcal{T}_{2\text{ph}}) = 2^{-(2d+1)} + (1 - 2^{-(2d+1)})2^{1-2d} = 2^{-2d} \left(\frac{5}{2} - 2^{-2d} \right) = 5 \cdot 2^{-(2d+1)} (1 + o(1)). \quad \square$$

Remark 7.2 (the simulator artifact; retraction of the exact law $1 - (1 - f)2^{-(2d+1)}$). The session’s simulators certified a pigeon as free when its row XOR equals 1 (odd count of 1s), after pre-checking, with oracle knowledge of ρ , that the pigeon is free. No legitimate degree-1 query implements that certification: by linearity (the mechanism of Lemma 3.1; explicitly Lemma 8.2 below) the stated query $Q_i = 1 + \sum_j x_{ij}$ answers 1 iff the row XOR is 0, the constant-free row sum answers 1 on *every* killed pigeon, and the oracle pre-check conditions on the very quantity (freedom) that the tree is supposed to be certifying. The exact law $1 - (1 - f)2^{-(2d+1)}$, with its “phase 2 never fails” step, is accordingly **retracted**. The previously recorded measurements — success 0.970 at $(32, 2)$, 0.97–0.98 at $(64, 2)$, 0.974 at $(128, 2)$, 0.998 at $(64, 4)$, and the five-point replication over 816,000 simulations — verify the *hybrid* law of the artifact simulator (certify on “free and row-XOR odd”), not the tree; the replication’s $(48, 4)$ fluctuation investigation stands, but within the artifact. Table 2 separates the hybrid rows from the literal rows, the latter confirming Theorem 7.1 point for point, including by exact enumeration.

Remark 7.3 (retry variant, literal law). If phase 1 collects the full certified set C (the free pigeons whose row answered even, $|C| \sim \text{Bin}(2d + 1, \frac{1}{2})$) and phase 2 retries across C , the rows being disjoint, the failure event is “every certified row all-zero”, of probability

$$\left(\frac{1 + 2^{1-2d}}{2} \right)^{2d+1} = 2^{-(2d+1)} (1 + o(1)) :$$

the retry recovers the retracted law’s asymptotic constant. Measured (literal re-implementation): success 0.9430 at $(32, 2)$ against 0.9448 predicted; exact enumeration at $(5, 2)$: strict success $0.9436865 = 1 - (9/16)^5$, digit-exact. The earlier retry measurements (0.975/0.968/0.970 at $(n, d) = (32, 2), (64, 2), (128, 2)$, 0.9925 and 1.0000 at $d = 4$, 1.0000 at $d = 8$) were produced by the artifact simulator and verify the hybrid law (Remark 7.2).

7.2 Model fidelity: the rigidity against the literal pipeline

Theorem 7.1 is a theorem about the coin channel. Against the literal pipeline of Definition 2.1 the phase-1 certification channel does not exist.

Corollary 7.4 (the parity-certification channel is closed against the pipeline). *Against the literal pipeline of Definition 2.1 the two-phase tree’s phase 1 never certifies: every pigeon answers the determined 0 to its row sum (Proposition 3.4), phase 1 completes without firing, and the fallback decides, so the tree’s success collapses to the base rate f , far below either coin-channel value of Theorem 7.1 (hybrid or literal). Measured on the exact pipeline (uniform designs over $\text{Des}(4, 2)$) at $(32, 2)$, budget 1000: two-phase 0.2767 [0.2323, 0.3260] against the single-variable scan’s 0.2650 [0.2213, 0.3138] — statistically indistinguishable, the phase-1 signal void (0 answers 1 among 2000 free-pigeon row-sum queries). The “reversed tree” scenario of the previous draft (stop at the first 0, then scan the certified row for a 1, winning with probability exactly 1) does not exist: it required assigned pigeons to answer 1, which the zeroing restriction semantics excludes under either presentation (Remark 3.5); every pigeon answers 0, so a first 0 certifies nothing. The parity-certificate mechanism is dead at kernel level; the certificate mechanisms that survive against the pipeline are the counting certificates (Lemma 8.4, over budget) and the adjacency certificates (Section 9).*

Proof. By Proposition 3.4, Q_i^p is a restricted consequence for free pigeons and the zero polynomial for assigned pigeons, so every row-sum answer is the determined 0 and phase 1 never fires; the fallback labels an arbitrary pair, free with probability f . For the reversed scenario: the first 0 is answered by every pigeon, free or assigned, so it certifies nothing; with the assigned-row variables zeroed there is no reading under which an assigned pigeon answers 1. The measured values are the exact-pipeline rows of the chaining study (Section 9). $\square$

Remark 7.5 (what the fidelity means). Three lessons. First, the hybrid rows of Table 2 validate the artifact simulator’s certification law, and the literal rows validate Theorem 7.1: a sampler that draws free-pair design bits independently implements the coin channel, not the design space, whose row parities are forced (Proposition 3.4); the coin channel drops exactly the row-parity constraints (legal fraction $\approx 2^{-(2d+1)}$, Remark 3.5) and, at degree 2, the product identities (Remark 6.3). Second, the rigidity is good for the pseudo-solution program: the literal $\Omega(n, d)$ at $p = 2$ is strictly more robust against parity-certification attacks than the coin model suggests, and both the solution condition (Definition 2.2) and the chi-task hypothesis (Theorem 2.3) survive the strongest known adaptive attacker (Section 9). Third, aggregate queries are not fully dead in principle: a full scan of a row reveals its parity, and an assigned row always shows exactly one answer 1 while a free row shows an odd number of them (parity-locking, Theorem 3.2(iv)); at $d = 2$ a free row shows three 1s with probability $\frac{1}{2}$ (odd counts 1 and 3 being equally likely, $\binom{4}{c}/2^3$ on odd c), a detectable event. But each full scan costs $2d$ queries against a free-row density $\approx 2d/n$, so within budget $e = d \log k$ the detection probability is $O(ed/n^2)$ -small, consistent with the cap of Theorem 5.3. Quantifying the best statistical freeness signal obtainable from aggregate queries within budget is Open Problem 3.

8 The certification floor at degree 1

This section settles the unbounded-budget certification problem at degree 1. All trees below query single outer variables only and have no query budget; adaptivity is allowed throughout but will not help.

Definition 8.1 (answer table; configuration). The *answer table* is $M \in \mathbb{F}_2^{(n+1) \times n}$ with $M(i, j) = \text{ans}(x_{ij})$; call $M(i, j) = 1$ a *one-entry*. A *configuration* is a pair $(Z, c) = ((D, R, \mu), \text{coins})$ of the pipeline: the status data Z (the free sets D, R and the matching μ) together with the free-region coin draws c ; M is a function of the configuration. In this section the free cells' answers are the independent fair bits of $\text{Coin}(n, d)$ (Definition 3.7), the answer-table model that the session's exact-channel samplers implement and in which the closed forms below are verified; fidelity to the literal pipeline is discussed in Remark 8.10. Fallback convention: a tree that exhausts its search outputs a fixed fallback pair, free with the base rate $f = (2d + 1)2d/((n + 1)n)$; "strict" exact values below carry no fallback luck (the canonical fallback pair is matched).

Lemma 8.2 (linearity of answers). *For any affine-linear query $g = a + \sum_{p \in S} x_p$ ($a \in \mathbb{F}_2$, S any set of pairs: the entire degree-1 query class), $\text{ans}(g) = a + \bigoplus_{p \in S} \text{ans}(x_p)$.*

Proof. $\text{ans}(g) = L(g^\rho) = L(a \cdot 1 + \sum_p x_p^\rho) = aL(1) + \sum_p L(x_p^\rho) = a + \sum_p \text{ans}(x_p)$ over $\mathbb{F}_2$, using the linearity of L and the design axiom $L(1) = 1$. $\square$

Corollary 8.3 (full-scan dominance). *For every adaptive degree-1 tree T there is a non-adaptive tree T' that queries all $n(n+1)$ single variables and then outputs, with the same success probability as T on every configuration. Consequently the optimal error over the whole adaptive degree-1 family equals the Bayes error of the pair posterior given M ,*

$$\text{err}^* = \mathbb{E}_M \left[1 - \max_p \mathbb{P}[p \in F \mid M] \right],$$

and randomized trees do not beat the deterministic optimum (convexity).

Proof. By Lemma 8.2 the answer to any query of T is an $\mathbb{F}_2$ -function of M . By induction on the query index, T 's t -th query and its answer are functions of M for every t , so T 's output label is a function of M . T' computes M by a fixed full scan, then replays T 's decision procedure, answering each simulated query by Lemma 8.2, and outputs T 's final label. The displayed identity is the definition of the optimal measurable selector of a pair given M ; a randomized tree is a mixture of deterministic trees, so the optimum is attained by a deterministic one. $\square$

Lemma 8.4 (counting certificates). *In every configuration, every killed pigeon's row of M contains exactly one one-entry (at its matched hole) and every killed hole's column contains exactly one one-entry (at its matched pigeon). Hence, soundly against every configuration consistent with the transcript:*

- (a) a row containing any number of 1s other than exactly one certifies its pigeon free;
- (b) a column containing any number of 1s other than exactly one certifies its hole free;
- (c) a one-entry in the row of a certified-free pigeon certifies its hole free, and a one-entry in the column of a certified-free hole certifies its pigeon free.

Proof. Killed pigeon i : $x_{i,\mu(i)}$ answers 1 (matched, Theorem 3.2(i)); every other pair (i, j') of the row is killed–unmatched and answers 0 (Theorem 3.2(ii)); there is no free hole in the row because $i \notin D$. Symmetrically for columns. Soundness of (a): if the pigeon were killed, its row would show exactly one 1. Soundness of (c): with the pigeon certified free, a one-entry (i, j) cannot be matched, and killed j would make (i, j) killed–unmatched (answer 0). $\square$

The session’s earlier trees aggregate answers by XOR ($Q_i = 1 + \text{row XOR}$, Theorem 7.1). The certificates above need the *count*, which a decision tree may compute but no single parity query can express; this is the whole gap between Theorem 7.1 and the floor, and the mechanism that the parity framing missed.

Lemma 8.5 (pigeonhole rescue). *If every row of M shows exactly one 1, then some column shows at least two 1s (there are $n+1$ rows and n columns). In that event the column is certified free by Lemma 8.4(b), a certified-free column contains no matched pair, so every one-entry in it certifies its pigeon free by Lemma 8.4(c), and the tree outputs a certified free pair (error 0 on this event).*

Lemma 8.6 (the near-permutation class). *Define class $\mathbf{b} = \{\text{every column of } M \text{ shows exactly one 1, and every row shows at most one 1}\}$. Then:*

- (i) $\mathbb{P}[\mathbf{b}] = (2d+1)(2d)! / 2^{(2d+1)2d}$;
- (ii) *on class $\mathbf{b}$ the free-region coin matrix is a permutation matrix on $2d$ of the $2d+1$ free pigeons plus one all-zero row, and every one-entry of M lies in the near-permutation formed by the killed matching and that partial permutation;*
- (iii) *the configurations consistent with a class- $\mathbf{b}$ table are exactly the $\binom{n}{n-2d}$ choices of $n-2d$ one-entries as the matched pairs; each hole is free in a fraction exactly $2d/n$ of them, the all-zero row’s pigeon is free in all of them, and no pair has posterior above $2d/n$;*
- (iv) *the Bayes error on class $\mathbf{b}$ is exactly $1 - 2d/n$.*

Proof. (i) Killed rows and columns show exactly one 1 automatically, so the condition lives on the $(2d+1) \times 2d$ free-region coin matrix: each of the $2d$ columns has exactly one 1, and the rows sum to $2d$ with each row at most one 1, forcing exactly one all-zero row and a bijection elsewhere: $(2d+1)$ choices times $(2d)!$ permutation matrices, out of $2^{(2d+1)2d}$ equally likely coin matrices. (ii) Immediate. (iii) A consistent configuration is any choice of $n-2d$ one-entries as μ , with killed sets the endpoints of the chosen entries: disjoint rows and columns hold automatically (each column has one one-entry, rows at most one), every unchosen one-entry has both endpoints free, and the coins are forced; conversely every configuration generating this M arises so. Each hole’s column edge is chosen in $\binom{n-1}{n-2d-1}$ of the $\binom{n}{n-2d}$ twins, so $\mathbb{P}[\text{hole free}] = 1 - \binom{n-1}{n-2d-1} / \binom{n}{n-2d} = 2d/n$; the all-zero row cannot be matched, so its pigeon has posterior 1; every matched-candidate pigeon has posterior $1 - 2d/n$; and hence every pair has joint posterior at most $2d/n$ (pairs with the all-zero row: exactly $2d/n$; all other pairs: bounded by the hole’s $2d/n$). (iv) Immediate. $\square$

Theorem 8.7 (exact certification floor at degree 1). *Every adaptive degree-1 tree for the pipeline at $p = 2$, with no query budget, in the answer-table model of Definition 8.1, has error probability exactly*

$$\text{err}^* = \left(1 - \frac{2d}{n}\right) \cdot \frac{(2d+1)(2d)!}{2^{(2d+1)2d}} = 2^{-4d^2 + 2d \log_2 d - O(d)} = 2^{-\Theta(d^2)},$$

and the following non-adaptive tree attains it: query all $n(n+1)$ variables in fixed order; compute all row and column counts; then

- (1) if some row and some column both have count $\neq 1$, output (that row's pigeon, that column's hole);
- (2) else if every row has count exactly 1, output the first one-entry of the first column with count ≥ 2 ;
- (3) else if every column has count exactly 1 and some row has count ≥ 2 , output that pigeon with its first 1-column;
- (4) else (class **b**) output the count-0 row's pigeon with any hole.

Proof. The four cases exhaust all M . Cases 1–3 output pairs with posterior 1 by Lemmas 8.4 and 8.5, the best possible. Case 4 is class **b**, where by Lemma 8.6 the posterior of every pair is at most $2d/n$ and the tree attains it. By Corollary 8.3 no tree exceeds the Bayes value on any M , so the displayed value is the exact optimum. The asymptotic form is Stirling's applied to $\log_2 \text{err}^* = -(2d+1)2d + \log_2((2d+1)(2d)!) + O(1)$. $\square$

At $(32, 2)$: $\text{err}^* = 1.0 \times 10^{-4}$, against the conjectured floor $2^{-5} = 3.1 \times 10^{-2}$ of the earlier draft and Theorem 7.1's literal error $39/256 = 1.5 \times 10^{-1}$; at $(64, 4)$: $\text{err}^* = 6.7 \times 10^{-17}$ against 2^{-9} . The $2^{-\Theta(d)}$ certification-floor conjecture and the two-phase tree's optimality are both false.

Remark 8.8 (verification record). All closed forms are verified digit-exactly by exhaustive computation in one registered run of `cert_floor_check.py` (31s; fixed seeds, registered output reproduced in the corpus).

1. *Bayes audit at $(4, 1)$, exhaustive.* All $120 \times 64 = 7680$ configurations enumerated and grouped into 6120 answer-table classes. On every class the counting tree's output pair achieves exactly the maximum pair posterior ($\max |\text{tree} - \text{best}| = 0$), and the pooled Bayes error $\mathbb{E}_M[1 - \max \text{posterior}] = 0.046875$ equals $\text{err}^*(4, 1)$ exactly.
2. *Exact enumeration at $(5, 2)$.* All 2^{20} coin matrices: strict success $0.9998856 = 1 - 120/2^{20}$, matching $\mathbb{P}[\mathbf{b}] = (2d+1)(2d)!/2^{(2d+1)2d}$ digit for digit.
3. *Monte Carlo.* Counting-tree success 1.0000 at $(32, 2)$ in 2000 runs (expected failures 0.2 at $\text{err}^* = 1.0 \times 10^{-4}$), 1.0000 at $(64, 2)$ in 1500 runs, and 1.0000 at $(64, 4)$ in 800 runs ($\text{err}^* = 6.7 \times 10^{-17}$).

In a 200,000-configuration sample at $(5, 2)$, seven multi-configuration classes showed a nonzero tree-versus-Bayes gap; every one is a class-**b** table, where Lemma 8.6(iii) makes all pair posteriors flat at $2d/n$ and small samples fluctuate. A gap on any other table would have refuted Theorem 8.7; none occurred.

Proposition 8.9 (rows-and-columns parity tree, $O(n)$ queries). *Scan the $n+1$ row sums Q_i and the n column duals $K_j = 1 + \sum_i x_{ij}$ ($2n+1$ queries); let I be the set of pigeons that answered 1 and J the set of holes that answered 1 (both certified free, soundly: assigned lines answer the determined 0). Output $(\min I, \min J)$ if both are nonempty; if I is empty, the global parity identity (the sum of the row XORs equals the sum of the column XORs) forces some column to*

be certified, and the tree scans the certified columns for a one-entry and outputs it; symmetrically if J is empty. Its error is exactly

$$\text{err}_R = \frac{\sum_{b \text{ odd}} \binom{2d}{b} 2^{2d(2d-b-1)} + \sum_{a \text{ odd}} \binom{2d+1}{a} 2^{(2d-a)(2d-1)}}{2^{(2d+1)2d}} = (6d+2) 2^{-6d} (1 + o(1)),$$

versus Theorem 7.1's $5 \cdot 2^{-(2d+1)}$: an exponential gap of $2^{4d}/\text{poly}(d)$ at equal query cost. The purely non-adaptive variant (output $(\min I, \min J)$ only when both sets are nonempty, fallback otherwise) has error $2^{-(2d+1)} + 2^{-2d}$. Exact enumeration at $(5, 2)$: strict success $0.9965019 = 1 - 3668/2^{20}$ (the two failure forms contributing 1028 and 2640), digit-exact; measured 0.9955 at $(32, 2)$, 0.9960 at $(64, 2)$, 1.0000 at $(64, 4)$.

Proof. Certification is sound: assigned lines answer the determined 0, so an answer 1 certifies freeness. The non-adaptive variant errs iff no row or no column is certified, events of probability $2^{-(2d+1)}$ and 2^{-2d} ; they are disjoint, because both failing forces the sum of the row XORs and the sum of the column XORs to differ ($n+1$ odd versus n odd), contradicting the global parity identity. For the adaptive tree, the rescue branches leave two failure forms, F1 (no row certified and the column rescue failing: $\sum_{b \text{ odd}} \binom{2d}{b} 2^{2d(2d-b-1)}$ configurations) and F2 (rows certified but no column and the row rescue failing: $\sum_{a \text{ odd}} \binom{2d+1}{a} 2^{(2d-a)(2d-1)}$); dividing by $2^{(2d+1)2d}$ and applying Stirling gives the displayed asymptotics. The case analysis is mechanical; the closed form is confirmed digit-exactly by the exact enumeration quoted. $\square$

Remark 8.10 (floor fidelity: coin model versus literal pipeline). Theorem 8.7 is proved and verified in the answer-table model of Definition 8.1, in which the free cells' answers are independent fair bits. The full scan on which the counting tree relies leaves the regime of Corollary 3.8: it pins every row in its entirety, the event E_{full} on which the coin-channel equivalence for single-variable trees fails, and Proposition 3.4's forced row parities then constrain the literal pipeline's answer table (no free row of the literal pipeline is ever all-zero). A verbatim transfer of the floor to the literal pipeline's answer table is therefore not claimed here; the chi-transfer note of the corpus (for the counting tree the chi-indicator coincides with its error indicator whenever no degree- $\leq d$ consequence in $V(n, d)^\rho$ has constant term 1, a reading of Definition 2.1 to verify against [8]) transfers the floor to the chi-task directly under that reading. The budgeted floor, which is what Theorem 2.3 actually quantifies over, is Open Problem 3.

Remark 8.11 (Theorem 8.7 under parity-locking). The exact constant $\text{err}^* = (1 - 2d/n)(2d+1)(2d)!/2^{(2d+1)2d}$ is a *coin-model* number: it is proved and verified in the answer-table model of Definition 8.1, whose free rows are unconstrained. On the literal pipeline every free row is parity-locked (Theorem 3.2(iv)): a fully scanned free row shows an odd count of 1s, with $\mathbb{P}[\text{count} = c] = \binom{2d}{c}/2^{2d-1}$ on odd c — so the count-1 probability is $2d/2^{2d-1}$, exactly twice the coin model's $2d/2^{2d}$ — while killed rows still show exactly one 1. The counting mechanism survives (a count other than 1 still certifies freeness soundly, Lemma 8.4), but the hard class does not merely shift: it *vanishes*. Class **b** (Lemma 8.6) would require an all-zero free row: with $2d+1$ free rows and $2d$ free holes each showing exactly one 1, only $2d$ rows can carry a 1 — and parity-locking (forced row XOR 1) forbids exactly the all-zero row that **b** needs. Hence on the literal pipeline the full-scan counting tree succeeds *with probability exactly* 1: the unbounded floor terminates at zero, not at a changed constant (verified: 0 failures in 5000 exact-channel simulations). Theorem 8.7 therefore survives as the exact floor of the *coin channel* — the row-incomplete completion model, which is the correct model of the pipeline below per-row query

cost $\Theta(n)$ — and the open remnant is the *budgeted* all-degrees floor (Open Problem 3), whose degree- ≤ 2 form is settled at $\text{err}^*(d, d \log k) = k^{-\Theta(d^2/n)}$.

9 Adjacency certificates and the confirmation chain

The caps proved so far bound what a single answer can say: the per-hit posterior of any scan hit is q (Lemma 5.1, Theorem 6.1), and the measured confirmations show every single-hit strategy landing on q . This section exhibits the mechanism that defeats the per-query caps at the tree level. It rests on a certificate that consumes only the determined answers (Theorem 3.2(i)–(ii)), so it is sound against the literal pipeline of Definition 2.1 directly, with no coin-channel intermediate step and no property of the design values at all.

Lemma 9.1 (adjacency certification). *Let $p \neq p'$ be pairs sharing a row or a column. If $\text{ans}(x_p) = \text{ans}(x_{p'}) = 1$, then p and p' are both free. The certificate is error-free, and it is valid against the literal pipeline (Remark 3.5’s single design space).*

Proof. An answer 1 excludes the killed–unmatched status, which answers the determined 0 (Theorem 3.2(ii)); so each of p, p' is free or matched. A matched pair forces all of its row-neighbors and column-neighbors to the determined 0: if $\rho(i) = j$, then every other cell of row i is killed–unmatched, and so is every other cell of column j (the hole j is taken, and ρ is injective). Suppose $p' = (i, j)$ were matched; then p , being one of its row- or column-neighbors, would answer the determined 0 — contradicting $\text{ans}(x_p) = 1$. So p' is free. Symmetrically, if p were matched, its neighbor p' would be killed–unmatched and answer 0; and p itself is not killed–unmatched; hence p is free. Every step is a ρ -determined answer law, which no design can alter. $\square$

9.1 The confirmation chain and its measurements

The *confirmation chain* (`confirm` in the session’s chaining study) spends a budget B of single-variable reads as follows: scan fresh pairs in random order; on a hit at p (answer 1), spend up to $k = 6$ reads on fresh row- and column-neighbors of p ; any neighbor answering 1 certifies p free by Lemma 9.1 and is output immediately; otherwise resume scanning; at budget end, output the least-confirmed accumulated hit. The certificate events are what the caps cannot price: Lemma 9.1 makes their posterior exactly 1, so the chain’s success is the probability of *reaching* a certainty event, a quantity that grows with budget, whereas every previously capped quantity was budget-independent.

Remark 9.2 (measured chain performance). Registered run (`chi_and_chain.py`: 600 simulations per cell, Wilson 99% confidence intervals, budget-honest accounting; the self-check “every certified output is free” was asserted per simulation and never fired in 52,800 simulations).

1. *Stipulated coin channel.* Success 0.9317 [0.9001, 0.9538] at budget $B = 1000$ at both (32, 2) and (64, 4), against the single-variable scan’s 0.2550 [0.2120, 0.3033] and 0.3800 [0.3305, 0.4321] (a $3.7\times$ lift at (32, 2)); the confidence intervals are disjoint from the scan’s at every $B \geq 300$ at both sizes and overlap at $B = 100$. Posterior given certification 1.0000 in every cell. The certificate rate is exponential in budget: 5.17, 20.50, 49.17, 91.83% at $B = 30, 100, 300, 1000$ and (32, 2); 7.50, 20.83, 50.17, 91.83% at (64, 4); i.e. $-\ln(1 - \text{cert})/B$ lies in [0.0018, 0.0026], roughly constant at ≈ 0.0025 per query, so $\text{cert} \approx 1 - e^{-B/400}$.

2. *Exact pipeline* (uniform designs over $\text{Des}(4, 2)$, the one-kernel space of Remark 3.5): success 0.9417 [0.9119, 0.9618] at $(32, 2)$, $B = 1000$; certificate rate 92.83%; posterior given certification 1.0000 (asserted). The chain consumes only the ρ -determined facts (matched restricts to 1, killed to 0) and the marginal fairness of the free design bits (measured 0.4918 over $n = 8000$ samples), so nothing in its mechanism changes under parity-locking (odd row parities permit the neighbor 1s); it is the only tested strategy that beats the single-variable cap on the true pipeline at $(32, 2)$.
3. *The per-hit no-lift is intact*. The chain’s posterior given a plain scan hit falls *below* q , 0.1633 [0.0696, 0.3372]: free hits self-select into certifications, leaving a matched-enriched residue. The lift is not a better guess but a certainty reached more often; the single-pass no-lift (Theorem 6.1) is sharp, and the cap it proves is a per-query, not a per-tree, bound.
4. *No lift without adjacency*. Budget-honest AND-scans are worse than single variables (0.1333 against 0.2550 at $(32, 2)$, $B = 1000$), and AND-pool and AND-split strategies sit at the cap (0.2317 and 0.2717, overlapping the scan’s interval); a posterior-weighted chain that pools certified neighbors with plain hits lifts by the same mechanism, diluted (0.3483 at $(32, 2)$ and 0.8000 at $(64, 4)$, $B = 1000$). The earlier “no lift found at toy scale” reading is superseded: the probe that supported it never implemented its product-test branch (a measurement bug recorded in the corpus).

Remark 9.3 (compatibility with the proved caps). Lemma 9.1 does not contradict Theorem 5.3 (or the no-lift theorem): the coupling slack of the cap is budget-dependent and vacuous at $e \asymp 10^3$ ($e/n \approx 31$ at $(32, 2)$, $B = 1000$), while at the program’s budget $e' = d \log k \ll n$ the slack is $o(1)$. What the chain changes is the shape of the budgeted question: its error decays in *budget* like $e^{-B/400}$ rather than staying at q , so the printed chi-hypothesis survives or falls with the scaling of the certificate constant in (n, d) (Open Problem 4).

Remark 9.4 (place in the certificate trichotomy). The known freeness certificates divide three ways. *Parity* certificates — query the row sum, certify on answer 1 — are dead against the literal pipeline: Q_i answers the determined 0 on every pigeon (Proposition 3.4; measured 0/2000 on free pigeons), and the two-phase tree collapses to the base rate (Corollary 7.4). *Counting* certificates — line counts $\neq 1$, Lemma 8.4 — are alive and Bayes-optimal in the coin model unbounded (Theorem 8.7), but inert within any polylog budget (Section 10), and their exact constant is a coin-model number (Remark 8.11). *Adjacency* certificates (Lemma 9.1) are alive, error-free, sound on the pipeline, and the only mechanism measured to beat the per-query caps there; the budget scaling of their constant is the sharpest open problem of the session (Open Problem 4).

10 Budget reconciliation

Theorem 2.3 (Theorem 6.1 of [8]) quantifies over *budgeted* trees: (d, e') -trees with $e' = d \log k$ queries (the printed budget is $e' = O(\log n) + O(d \log n)$; we keep the shorthand $d \log k$). Every attacker constructed in this paper is over that budget in the intended regime $e' = \text{poly}(\log n) \ll n$:

attacker	queries	error (coin channel; chain: measured)
two-phase tree (Theorem 7.1)	$\asymp 2n + 1$	$5 \cdot 2^{-(2d+1)}$
rc parity tree (Proposition 8.9)	$2n + 1$	$(6d + 2) 2^{-6d}$
full-scan counting tree (Theorem 8.7)	$n(n + 1)$	$2^{-\Theta(d^2)}$
confirmation chain (Lem. 9.1)	B (single variables)	$\approx e^{-B/400}$, decaying in B

The first three errors are coin-channel closed forms (for the counting value the $2^{-\Theta(d^2)}$ order is kernel-faithful, Remark 8.11); against the literal pipeline the first two are dead outright (Corollary 7.4). The chain’s error is measured on both channels (Remark 9.2) and is the first to decay in budget rather than in d . The two-phase witness alone exceeds the budget $d \log k$ whenever $d < 2n^{1-\delta}$ at $k = 2^{n^\delta}$, *a fortiori* the counting tree. Counting certificates are inert within budget: a single variable answers 1 with probability $m + f/2 = O(1/n)$, so e single-variable queries elicit an expected number of 1s of order e/n , and a line-count certificate (two 1s in one row or column) does not occur until $e \gtrsim n$, far above $d \log k$. Within budget the certificate mechanism of Lemma 8.4 cannot fire, the cap of Theorem 5.3 governs (its slack $O(e/n)$ is $o(1)$ there), and the chi-hypothesis survives every known attack at the printed budget. The one attacker whose measured error decays in budget, the confirmation chain, needs $\approx c^{-1} \log(1/\gamma)$ queries with $c \approx 1/400$ per query at the measured points; whether $e' = d \log k$ suffices at the program’s parameters is the adjacency-budget race (Open Problem 4). Three precise statements record where matters stand:

1. The unbounded degree-1 quantifier is settled, by Theorem 8.7. The unbounded quantifier is not the printed one: Definition 2.2 quantifies over (d, e) -trees, and whether any reading of [8] leaves the class unbounded is a reading question to verify against the source. Under an unbounded reading of the solution condition, the counting tree’s error $2^{-\Theta(d^2)}$ falls below the required $\gamma = k^{-O(1)}$ precisely when $4d^2 - O(d \log d) > O(\log k)$, which holds throughout the program’s intended regime $d = (\log k)^{O(l)}$ with $l \geq 1$ fixed: the budgeted quantifier of Definition 2.2 is essential for the program’s viability at $p = 2$.
2. The budgeted floor is open: the optimal error of e -query adaptive degree-1 trees as a function of e and, in particular, whether any $e = d \log k$ tree beats the cap of Theorem 5.3 (Open Problem 3). The confirmation chain answers the large- e end: it beats every per-query cap for $B \geq 300$ at the measured points, with error decaying like $e^{-B/400}$ (Section 9); the open span is the scaling in between (Open Problems 3 and 4).
3. The chi-hypothesis of Theorem 2.3 remains open in exactly its printed budgeted form (Open Problem 1).

11 Consequences for the chi-task

Collecting the proved caps, at $p = 2$, on the pipeline $\Omega(n, d)$:

1. *Non-adaptive variable/monomial-class trees* (Proposition 4.6): success $\leq (\gamma s + 1)f + o(1)$; the chi-hypothesis is safe within the class up to $d = \Theta_\gamma(n^{2/3}(\log k)^{-1/3})$ at budget $s = d \log k$. The all-queries generality previously claimed for this cap is retracted (Remark 4.7).
2. *All adaptive single-variable trees within budget* (Theorem 5.3, Remark 5.4): success $\leq \max(q, f) + o(1)$; the hypothesis is safe with large margin for all $2d \leq \sqrt{2n}$, i.e. for $k = 2^{n^\delta}$ with $\delta \lesssim 1/(2O(l))$ under the program’s parameter connection $d = (\log k)^{O(l)}$. Without budget the same class is settled exactly at the Bayes floor $2^{-\Theta(d^2)}$ in the coin model (Theorem 8.7); the floor’s order is kernel-faithful and its constant is not (Remark 8.11).
3. *Single-pass degree-2 product tests* (Theorem 6.1): no per-hit posterior lift; sharp, and measured on the exact pipeline (Remark 6.3). The tree-level lift goes through adjacency and needs only single variables (Section 9).

4. *Certification-style attacks and the certificate trichotomy*: in the coin model the two-phase tree errs $5 \cdot 2^{-(2d+1)}$ under the literal query semantics (Theorem 7.1); it is dominated at equal $O(n)$ cost by the rc parity tree (Proposition 8.9) and, at $n(n+1)$ queries, by the counting tree’s Bayes-optimal $2^{-\Theta(d^2)}$ (Theorem 8.7). Against the literal pipeline the parity mechanism is dead outright (Corollary 7.4), the counting mechanism survives with a parity-locked hard class (Remark 8.11), and the adjacency mechanism is alive and measured: the confirmation chain reaches 0.9417 on the exact pipeline at $(32, 2)$, budget 1000 (Section 9). Every unbounded attacker is over budget, counting certificates are inert below $e \asymp n$ (Section 10), and the chain’s reach at the printed budget is exactly the adjacency-budget race (Open Problem 4). Hence the candidate $\Omega(n, 2)$ remains consistent with all known *budgeted* attackers at $p = 2$, and both the solution condition and the chi-hypothesis remain open and plausible in their printed budgeted forms.
5. *Trivial multiplicative tests do not refute the candidate*: an adaptive degree-2 tree built from random monomial triples detects violations of multiplicativity of $\omega = L \circ \rho$ outside the free region with measured probability 5×10^{-4} over the full pipeline (4000 samples), against 0.37 for a canonical computation that drops the restriction layer: ω is automatically multiplicative outside the tiny free region, whose all-free-triple density is $\Theta((d/n)^4) \rightarrow 0$.

The genuinely open classes are: budgeted degree-1 trees at intermediate budgets (the budgeted floor, with the confirmation chain as its strongest measured witness), and adaptive trees of degree ≥ 2 beyond the adjacency mechanism; Section 12 formalizes the remaining problems.

11.1 Theorem-faithfulness

The coin channel $\text{Coin}(n, d)$ and the literal pipeline agree *exactly* (total variation 0) on the marginal and joint single-variable laws of every row-incomplete query set, and disagree on a fully queried free row (conditional total variation $\frac{1}{2}$; parity-locking, Theorem 3.2(iv)) and at degree 2 (Remark 6.3). Table 3 records the status of each result; “kernel-faithful” means the proved statement holds verbatim for Definition 2.1’s pipeline.

[†]Measured-consistent: true monomial queries on the exact pipeline land on the cap, 0.2629 [0.2129, 0.3199] against $q = 0.2632$ (Remark 6.3).

12 Open problems

Open Problem 1 (the chi-hypothesis, printed budgeted form). Is there a constant C such that every (d, e') -tree with $e' = d \log k$ (printed: $e' = O(\log n) + O(d \log n)$) and queries of degree $\leq d$ has, on the pipeline $\Omega(n, d)$ at $p = 2$ with $d = (\log k)^{O(l)} \leq \sqrt{n}$,

$$\mathbb{P}[\text{failure}] \geq k^{-C} ?$$

Equivalently: no shallow low-degree adaptive tree certifies freeness with probability $1 - k^{-O(1)}$. An affirmative answer, together with Theorem 2.3, yields k -step lower bounds for $\text{AC}^0[l](\text{MOD}_2)$ -Frege refutations of $\neg\text{PHP}_n$ with $k = 2^{n^\delta}$, $\delta \approx 1/(2O(l))$: the first super-polynomial $\text{AC}[2]$ -Frege lower bounds, on the pigeonhole principle. A negative answer would refute $\Omega(n, d)$ as a pseudo-solution at $p = 2$ and redirect the program. Proved here: the affirmative answer for non-adaptive variable/monomial-class trees (Proposition 4.6) and for adaptive single-variable trees at budget

result	status	why
coin equivalence (Cor. 3.8)	kernel-faithful	exact off E_{full} ($\text{TV} = 0$); $\frac{1}{2}$ on full rows
channel law (Thm. 3.2)	kernel-faithful	it <i>is</i> the kernel law (verified)
completion (Lem. 3.6)	kernel-faithful	the zeroing completion, all d
Prop. 4.3 (A), 4.4 (C)	kernel-faithful	pair symmetry; single variables
Prop. 4.6 (D, variable/monomial class)	kernel-faithful	Lemma 4.1 + depletion
Thm. 5.3, Bayes caps (Lem. 5.1)	kernel-faithful	single variables; $\text{TV} = 0$ off E_{full}
counting certificates (Lem. 8.4)	kernel-faithful	determined answers; odd-count law corrected
Thm. 8.7's <i>order</i> $2^{-\Theta(d^2)}$	kernel-faithful	Rem. 8.11
adjacency certificate (Lem. 9.1)	kernel-faithful	determined answers only
two-phase law (Thm. 7.1)	coin-model-only	parity mechanism dead (Cor. 7.4)
rc parity tree (Prop. 8.9)	coin-model-only	row sums answer 0 (Prop. 3.4)
Thm. 8.7's <i>constant</i>	coin-model-only	hard class parity-locked (Rem. 8.11)
AND no-lift identity (Thm. 6.1)	coin-model-only [†]	product semantics; transfer unproved

Table 3: Theorem-faithfulness against the literal pipeline of Definition 2.1. “Kernel-faithful”: the statement holds verbatim for the pipeline (the coin channel agrees with it exactly on the query sets the result uses). “Coin-model-only”: the statement is proved in $\text{Coin}(n, d)$ and does not transfer verbatim; for Theorem 6.1 the pipeline transfer is measured-consistent but unproved. Results above the second rule are used in their kernel-faithful form throughout Sections 10 and 12.

(Theorem 5.3); without budget, degree 1 is settled exactly at the Bayes floor $2^{-\Theta(d^2)}$ (Theorem 8.7), which stays above $k^{-O(1)}$ only while $d^2 \lesssim \log k$ — one more reason the budgeted quantifier is the right one. The residual classes: budgeted degree-1 trees at intermediate budgets (Open Problem 3), now dominated in the measured range by the confirmation chain, whose error decays in budget like $e^{-B/400}$ on the pipeline itself (Section 9; its budget scaling is Open Problem 4), and adaptive trees of degree ≥ 2 beyond the adjacency mechanism (Open Problem 2).

Open Problem 2 (chaining beyond the per-hit posterior). The per-hit no-lift is sharp: every single-pass test has posterior exactly q (Theorem 6.1), measured on the exact pipeline as well (Remark 6.3). At the tree level the picture inverted: adaptive chaining *does* beat the per-query posterior cap — not by raising any single answer’s posterior, but by chaining a hit with an adjacent answer into a certain-free event whose probability grows exponentially with budget (Section 9: 0.9317 against the cap 0.2550 at $(32, 2)$, budget 1000, disjoint 99% intervals). The earlier “every tested chaining strategy decays as $n^{-1.8}$ to $n^{-1.9}$ ” reading is superseded; the probe behind it never implemented its product-test branch (a measurement bug recorded in the corpus). What remains open: (a) the budget scaling of the adjacency constant (Open Problem 4); (b) whether degree-2 chains add anything over the degree-1 confirmation chain — budget-honest AND-scans are *worse* than single variables, and AND-split chains certify far more rarely (4.00% versus 91.83% certificate rate at $(32, 2)$, budget 1000); (c) a cap theorem for the adaptive degree-1 class at intermediate budgets, between Theorem 5.3’s $e = o(n)$ regime and the unbounded floor (Theorem 8.7).

Open Problem 3 (the budgeted certification floor). What is the optimal error $\text{err}(e)$ of adaptive degree-1 trees with e queries, as a function of e ? Unbounded, the answer is exact: $\text{err}^* = 2^{-\Theta(d^2)}$, attained by the non-adaptive counting tree (Theorem 8.7); the earlier conjecture of a $2^{-\Theta(d)}$ floor, with the two-phase tree as its extremal witness, is false. The line-count certificates that attain the floor are inert below $e \asymp n$ (Section 10), and Theorem 5.3 caps the class at $\max(q, f)(1+o(1))$ for $e = o(n)$; the question is whether the truth interpolates or some $e = d \log k$

tree beats Theorem 5.3's cap, which is the content Theorem 2.3's hypothesis needs at $p = 2$. An affirmative budgeted floor would combine with Open Problem 1 to complete the $p = 2$ core of the $\text{AC}[2]$ -Frege program. Three sub-questions: (a) the exact Bayes program at degree 2 — Corollary 8.3's reduction dies there (a monomial answer is not an XOR of single answers), so no degree- ≥ 2 analogue of Theorem 8.7 is known; (b) against the literal pipeline, the best statistical freeness signal obtainable from aggregate (full-row or higher-degree) queries within budget, the quantitative remnant of Corollary 7.4; (c) the *parity-locked floor constant*: recompute Theorem 8.7's constant for the parity-locked answer table, in which every free row carries an odd count of 1s ($\mathbb{P}[\text{count} = 1] = 2d/2^{2d-1}$, twice the coin model's) and no free row is ever all-zero — the $2^{-\Theta(d^2)}$ order survives (Remark 8.11), the exact constant does not, and it is open.

Open Problem 4 (the adjacency-budget race). The confirmation chain's certificate rate is $\approx 1 - e^{-cB}$ at budget B , with $c \approx 1/400$ per query measured at both $(32, 2)$ and $(64, 4)$ (Remark 9.2); its error on the exact pipeline decays like $e^{-B/400}$. Determine $c(n, d)$. The two measured points have nearly equal free densities f and hit rates, so the scaling of c in (n, d) is untested; the free-density heuristic of the chaining study gives $c \sim d^2/n^2$ to d^3/n^3 (each read hits a free pair with probability $\approx f$, and a free hit then certifies within a few neighbor reads), under which the budget needed to push the error below γ scales as $(n^3/d^3) \log(1/\gamma)$: polynomial in n , polylogarithmic in $1/\gamma$. The stakes are exact: if $e' = d \log k$ queries suffice at the program's parameters $d = (\log k)^{O(l)}$, the printed chi-hypothesis of Theorem 2.3 is **false** and the $p = 2$ route via Theorem 2.3 closes; if c decays polynomially in n , the hypothesis survives with the confirmation chain as its strongest known witness. This is the sharpest open problem of the session: the budgeted certification floor must cover a trichotomy of certificate mechanisms — parity (dead on the pipeline, Corollary 7.4), counting (inert within polylog budget, Section 10), and adjacency (alive; this problem).

Open Problem 5 (the odd-characteristic analogue). Fix an odd prime p (any $p \geq 3$). The reduction chain of [8] is characteristic-uniform (Theorems 2.2, 3.2, 3.3, Definition 4.3, and Theorem 6.1 there are all stated for an arbitrary prime), and the free-answer alphabet grows to all of $\mathbb{F}_p$: killed lines still answer determinedly, free single variables answer uniform p -ary dies, so a fraction $\frac{p-2}{p}$ of free single-variable answers are self-certifying and the single-variable lane is open at odd p in a way it is not at $p = 2$. Determine whether, for every (d, e') -tree T' with $d = (\log k)^{O(l)}$ and $e' = O(\log n) + O(d \log n)$,

$$\mathbb{P}_{\rho, P}[\chi(P, \rho) = 1] \geq k^{-O(1)},$$

the hypothesis of Theorem 2.3 at characteristic p . A positive answer at any $k(n) \geq n^{\omega(1)}$ yields, via the Extended Nullstellensatz equivalence, k -step lower bounds for $\text{AC}^0[p]$ -Frege refutations of $\neg\text{PHP}_n$: the first super-polynomial lower bound for the full $\text{AC}^0[p]$ -Frege system at that characteristic, and the first lower bound of any kind for odd- p $\text{AC}^0[p]$ -Frege proofs of the pigeonhole principle; the odd- p parity-resolution lane is strictly less crowded (no dag-like $\text{Res}(\text{lin}_p)$ pigeonhole bound of any kind exists for odd p , only tree-like and fragment results). A negative answer (a tree with error $< k^{-O(1)}$) refutes $\Omega(n, d)$ at that p and redirects the program, exactly as at $p = 2$. Caveat: the p -ary two-phase tree's proposed error $p^{-(2d+1)}$ is flagged *analysis* in the corpus; it inherits the phase-2 coin-branch issue corrected in Theorem 7.1 (the certified row's sum being nonzero does not prevent the all-zero configuration), and its literal error needs the same recomputation before any use.

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